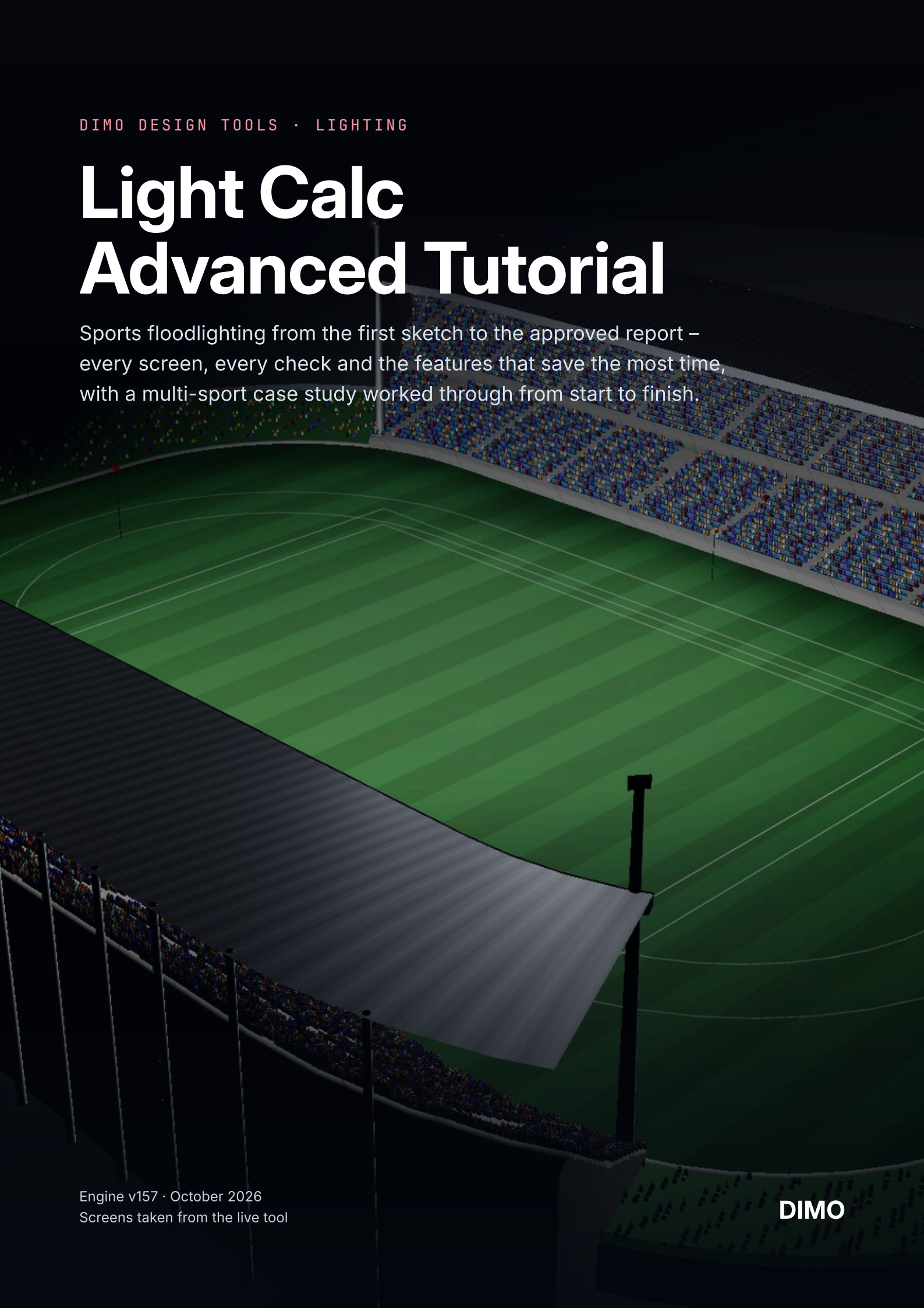
DIMO DESIGN TOOLS · LIGHTING

Light Calc Advanced Tutorial

Sports floodlighting from the first sketch to the approved report – every screen, every check and the features that save the most time, with a multi-sport case study worked through from start to finish.



Engine v157 · October 2026
Screens taken from the live tool

DIMO

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ABOUT THIS TUTORIAL

Every screen in it was taken from Light Calc engines v155–v157, working on real designs: a football ground at EN 12193 Class I in Part B and a fictional multi-sport stadium in Part D. Results are the tool’s own; figures such as luminaire colour data and the houses round the case-study ground are invented for the example and marked as such. The demo floodlights are generic – use the maker’s IES or LDT file for a real design.

PART A

Getting around

What Light Calc does, how the screen is laid out, and the three habits that keep work fast: calculate on demand, undo freely, let the tool keep the design.

CHAPTER 1

What Light Calc does

Light Calc designs and checks outdoor sports floodlighting – masts or roof-mounted – against the standard you choose, point by point from the luminaires' photometric files, and produces the report, aiming schedule and exports a contractor and an approving engineer need.

1
Sport & standard
Cricket, football, rugby, athletics, tennis, pickleball, volleyball, or several sports on one site

2
Ground
Standard dimensions, OpenStreetMap, or your own drawing traced to scale

3
Luminaires
IES / LDT files; several to compare or mix beams

4
Masts or roof
Placed by you or by the tool, within the mast rules; or a roof catwalk

5
Automatic design
Heights, numbers, aiming and optimising until every target passes

6
Scenes & DMX
Match, training, televised, security – groups, levels, maximum demand

7
Checks
TV cameras, glare, neighbours, stands, colour and flicker

8
Report & exports
PDF report and aiming schedule, Relux .rdf, DXF, electrical design

What makes it different

- **It designs, not only calculates.** Press one button and it places the masts, sets their heights and the number of floodlights, aims and optimises every floodlight, then trims masts and floodlights while every check still passes – for each luminaire you tick, alone and mixed, keeping the best.
- **Scenes are checked, not assumed.** Each scene – match, training, televised, security – is calculated with only its floodlights at their levels and checked against the class set for it.
- **The checks that approvals ask for are built in:** TV camera illuminance and uniformity, glare (CIE 112), obtrusive light to neighbours and the sky (CIE 150 / EN 12464-2), stands, colour and flicker for broadcast.
- **It is verified.** A validation suite of 21 test groups, including CIE 171:2006 case 5.6, runs against every engine version (chapter 20).
- **It connects.** Masts go to the electrical design (cables, distribution), to Poles & masts (shaft and foundation), and out to Relux and CAD.

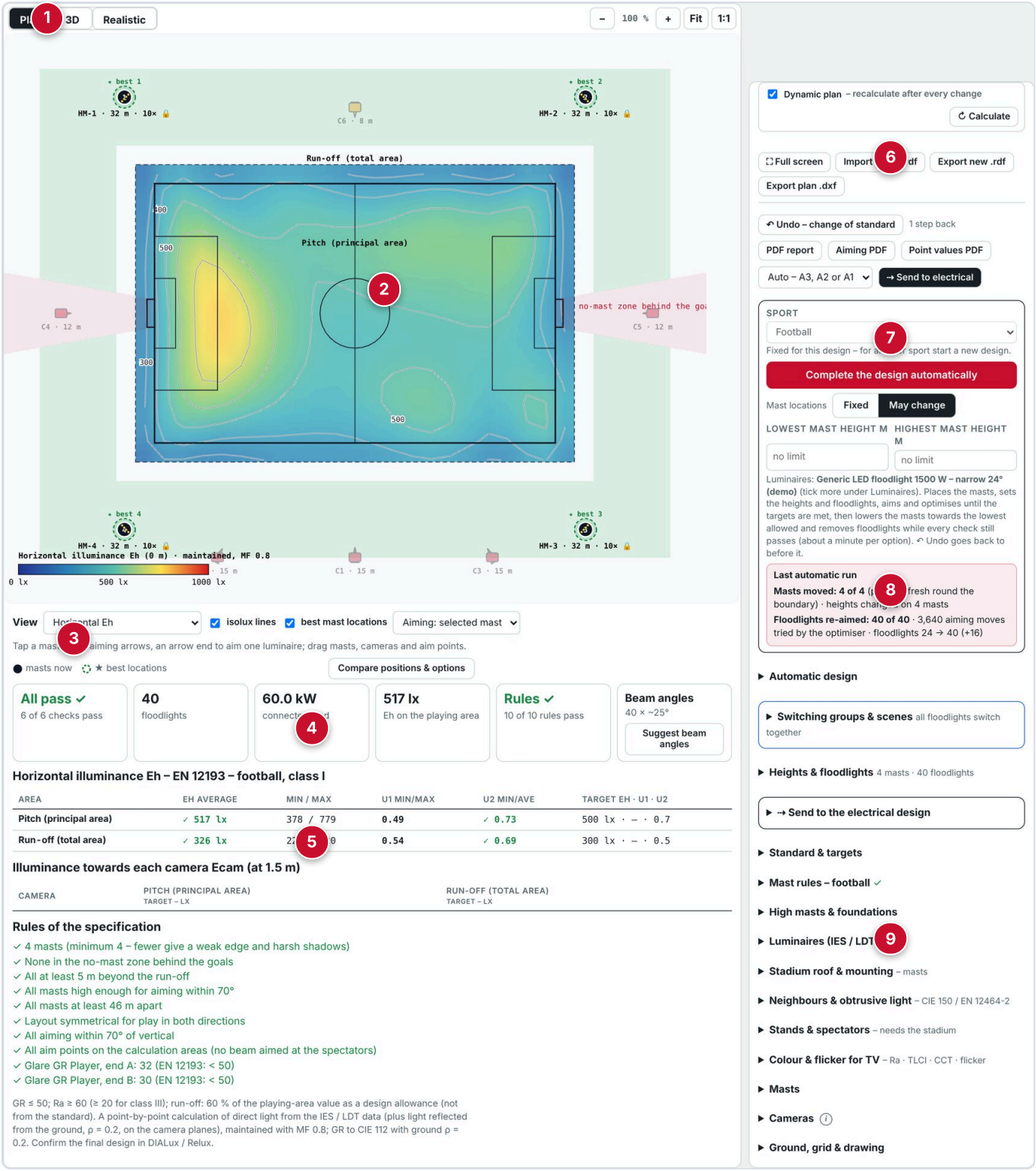
KEEP IN MIND

Light Calc is a design tool. Confirm a final design in DIALux or Relux where the client or standard asks for it – the .rdf export makes that a few minutes' work.

CHAPTER 2

The screen

Lighting → Lighting calc. The plan is on the left with the results under it; everything you set is in the side panel on the right, in panes you open as you need them.



Football, EN 12193 Class I, after the automatic design: 4 masts of 32 m, 40 floodlights, 60 kW, every check passing.

1 **Plan · 3D · Realistic** – the three views of the same design.

2 **The plan** – false colours, isolux lines, masts with height and floodlights, cameras, areas; drag to move, tap to select.

3. **View** – what the colours show (Eh, Ecam per camera, the lowest Ecam…), isolux lines, best mast locations, aiming arrows.
4. **Summary cards** – checks passing, floodlights, kW, average lux, rules, beam angles.
5. **Results** – Eh and Ecam tables against the targets, rules of the specification, glare.
6. **Toolbar** – full screen, Relux .rdf in/out, DXF, undo, PDF report, aiming and point-value PDFs, send to electrical.
7. **Sport card** – the sport and the big red **Complete the design automatically**.
8. **Last automatic run** – what the last run changed, in numbers.
9. **Panes** – Automatic design, Scenes, Heights, Standard, Mast rules, High masts, Luminaires, Stadium roof, Neighbours, Stands, Colour & flicker, Masts, Cameras, Ground.

CHAPTER 3

Calculating, undo and saving

Calculate on demand – or Dynamic plan

At the top of the side panel, **Dynamic plan** recalculates after every change. Untick it on a large design: changes then collect, a banner says *Not calculated since the last change*, and **Calculate the changes** works them all out at once. The calculation is incremental – moving one aim point recalculates one floodlight, not the design – so even with Dynamic plan on, aiming by hand stays quick.

Long runs (the automatic design, the optimiser) show their progress and a ■ **Stop** button. Stop during the automatic design puts the design back as it was before the run; Stop during the optimiser keeps the aim points moved so far.

Undo

↶ **Undo** in the toolbar names the step it will undo (*Undo – mounting*, *Undo – section type*) and counts the steps back – 30 are kept. The automatic design is one step: one Undo returns the design to before the run.

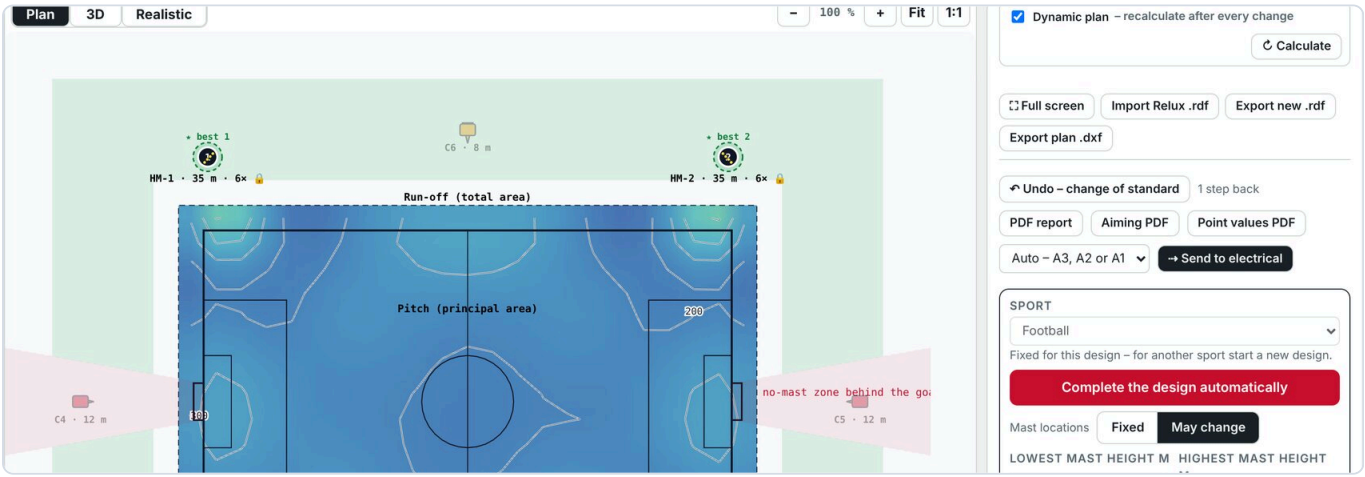
> NEW IN V155–V157
>
> Undo also covers the stadium roof, obtrusive light, stands and colour & flicker settings. Points under a pavilion or closed structure are no longer calculated. The design is locked while an automatic run is going.

Saving

The design is part of the project and is kept as you work – there is no Save button to forget. The PDF report carries the date and the design's revision; Relux .rdf and DXF exports are snapshots you can file with the submission.

Full screen and phones

⛶ **Full screen** gives the plan the whole window, with the side panel beside it. The tool also runs in the DIMO phone app: pinch to zoom the plan and the 3D views, tap to select, and files you export are saved through the app.



Before the first design: the football ground, the standard cameras and the green band where masts may stand, with ★ best locations.

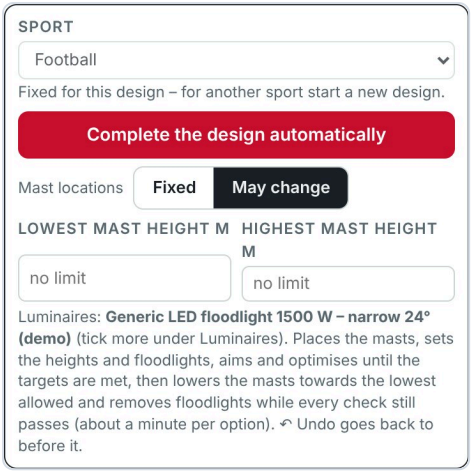
PART B

A football design, step by step

A full-size football pitch at EN 12193 Class I, from an empty ground to a passing design with scenes, DMX groups and the report. Each chapter covers one pane.

CHAPTER 4

Sport, standard and targets



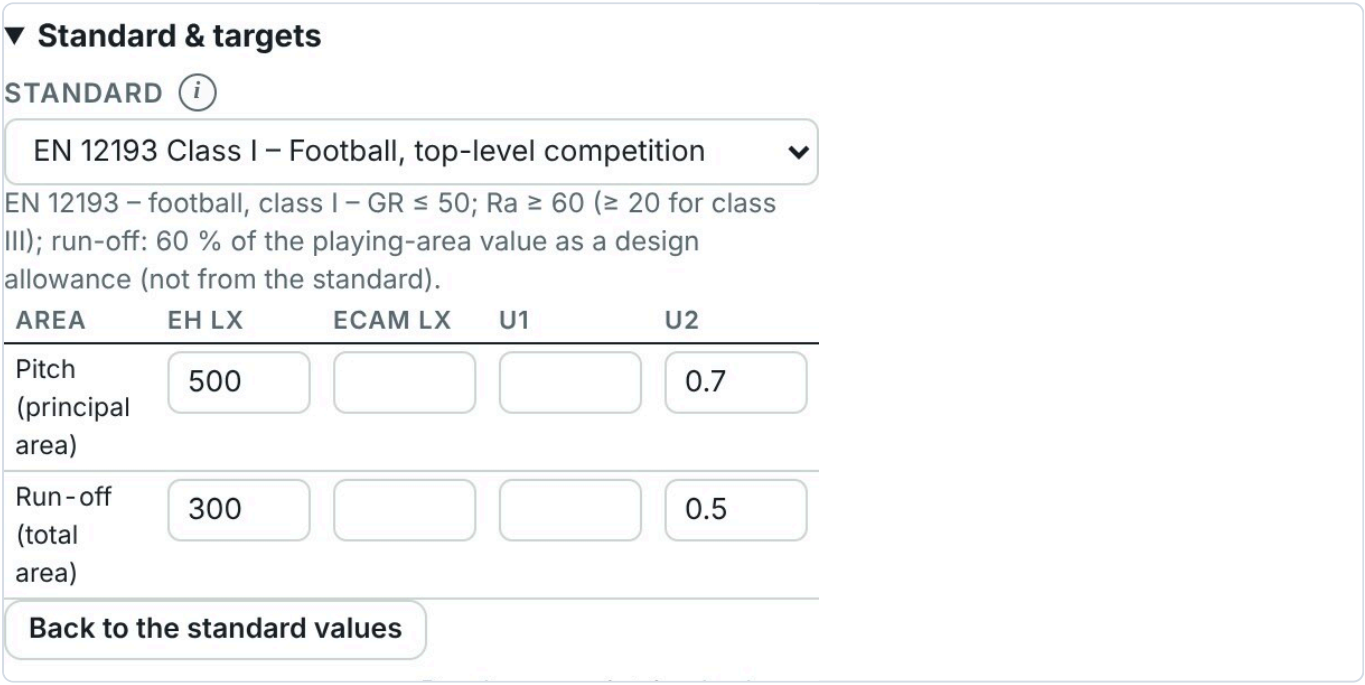
Sport card. Chosen first; fixed once the design starts.

Choose the sport first. It sets the ground, the calculation areas, the cameras, the mast rules and the standards on offer. Once the design starts the sport is fixed – another sport is a new design. For several sports on one site choose **Multi-sport facility** (Part D).

Mast locations – **May change** lets the automatic design place and move masts; **Fixed** keeps them where they are (existing masts, or masts fixed by the site). **Lowest / highest mast height** limit what the design may use – planning conditions often set a maximum.

Standards available for football

STANDARD	USE
EN 12193 Class I / II / III	top-level competition / club competition / training
Football – international broadcast (typical)	camera illuminance for TV
FIFA Class I – V	training … World-Cup-level televised



Standard & targets. The targets per area (Eh, Ecam, U1, U2) fill in from the standard and can be overtyped – **Back to the standard values** restores them. Below them the maintenance factor and ground reflection.

Maintenance factor

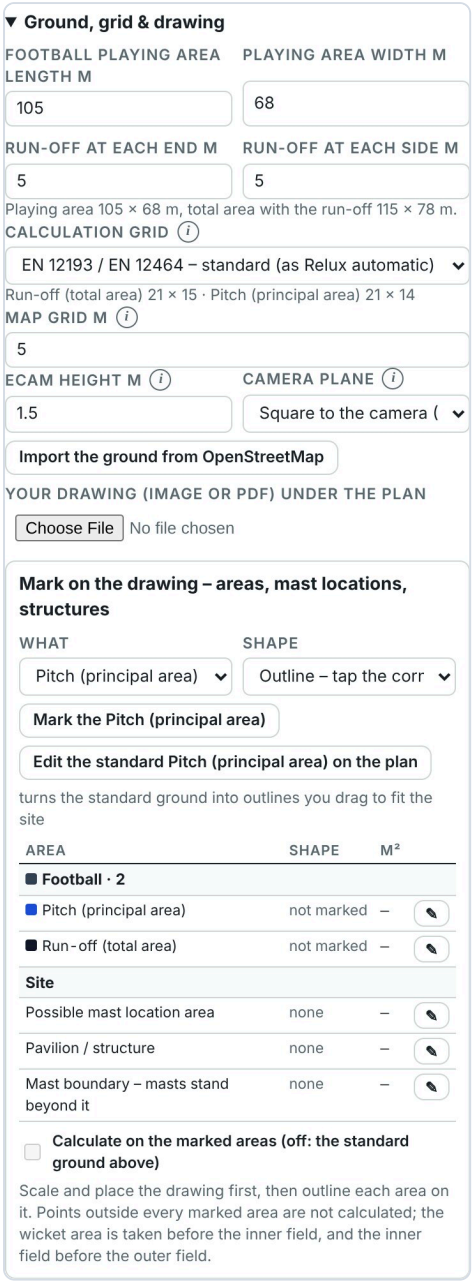
Results are maintained values – initial lux × MF. LED floodlights are usually 0.8 (L80 at end of life, clean optics); 0.7–0.75 for dusty sites or long cleaning intervals.

Light reflected from the ground

Tick to include it: grass reflects part of the light, which raises the vertical (camera) illuminance by roughly 10–15 % at ρ 0.20 but adds nothing to the horizontal. Leave it off for a conservative design; tick it to match a DIALux / Relux run that includes it.

CHAPTER 5

Ground, grid and drawing



Ground, grid & drawing.

The ground

Playing area and run-off for the sport, in metres. The calculation areas follow: for football the pitch (principal area) and the run-off (total area), each with its own targets.

Calculation grid

EN 12193 / EN 12464 – standard spaces the points as the standard asks (the same as Relux's automatic grid), so results compare one-to-one with a Relux check. A fixed spacing or your own point counts are also possible. **Map grid** sets the fineness of the colour plan only.

Cameras and Ecam

Ecam height (1.5 m usual) and **camera plane**: square to the camera (3D, as broadcasters measure) or vertical facing it.

Import the ground from OpenStreetMap

Type the ground's name or its latitude, longitude; Light Calc finds the outlines, you say which is the pitch and run-off, and it builds the areas, turned to north, with the map under the plan. Buildings round it can then be found for the obtrusive-light check (chapter 17). Step by step in chapter 31.

Your drawing under the plan

1. Load an image or PDF of the drawing.
2. **Set the scale: tap 2 points** a known distance apart and type the distance.
3. **Move the drawing** so its pitch lies on the pitch.
4. **Mark on the drawing** – outline the pitch, run-off, possible mast locations, pavilions and structures, a mast boundary.
5. Untick **Show the drawing** once traced – it stays with the design.

Worked cases: a PDF site plan in chapter 29, an aerial picture or Google Maps screenshot in chapter 30; every marking option in chapter 32.

Worked cases: a PDF site plan in chapter 29, an aerial picture or Google Maps screenshot in chapter 30; every marking option in chapter 32.

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Worked cases: a PDF site plan in chapter 29, an aerial picture or Google Maps screenshot in chapter 30; every marking option in chapter 32.

TIP

Mark **pavilions and structures** with their height and roof shape: they then block the light in the calculation, exactly – a roof between a floodlight and the pitch casts its real shadow.

CHAPTER 6

Luminaires

Upload IES or LDT files from the maker. Each luminaire shows its lumens and watts; **Use** makes it the default for new floodlights; ✕ removes it. Four generic demo floodlights (narrow, medium, wide, and a 400 W) let you try the tool before you have files.

The Auto tick – let the design compare

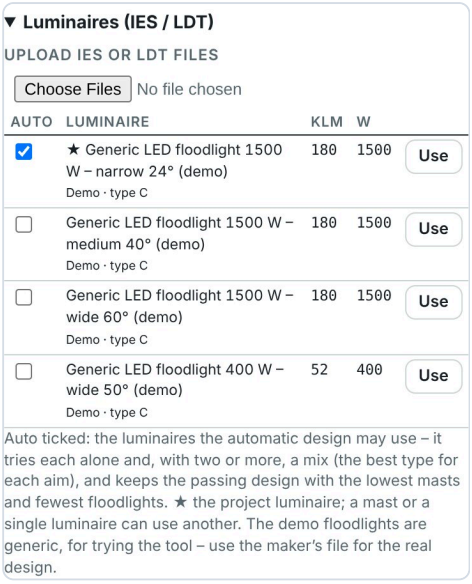
Tick every luminaire the automatic design may use. It designs with each one alone, then with all of them mixed, and keeps the passing option with the fewest floodlights and lowest masts. The others stay in memory: **Compare positions & options** shows each option's plan, results, floodlights and kW side by side.

Beam angles

The **Beam angles** card under the plan counts floodlights per beam; **Suggest beam angles** proposes narrow beams for far aim points and wider ones near the mast – mixing beams usually improves uniformity with the same count.

Colour and flicker data

CCT, Ra, R9, TLCI, flicker and frame rate per luminaire are entered under **Colour & flicker for TV** (chapter 19); CCT and Ra come in by themselves from LDT files.



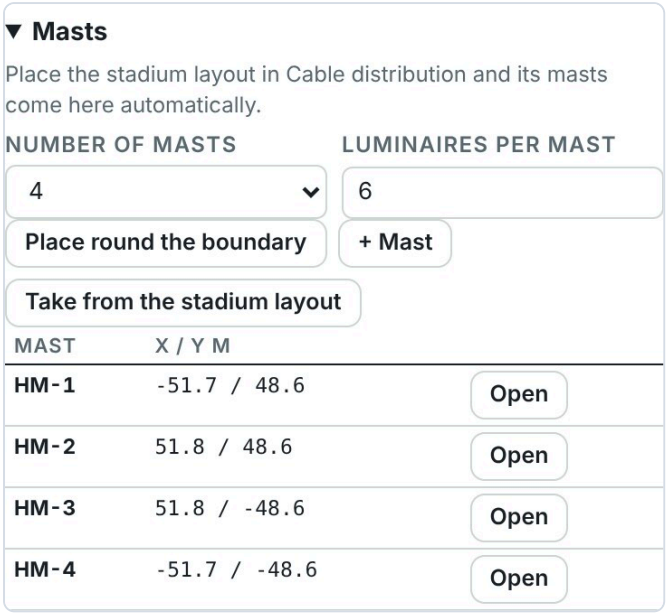
Luminaires. The ★ marks the default; ticks mark those the automatic design may use.

WATCH OUT

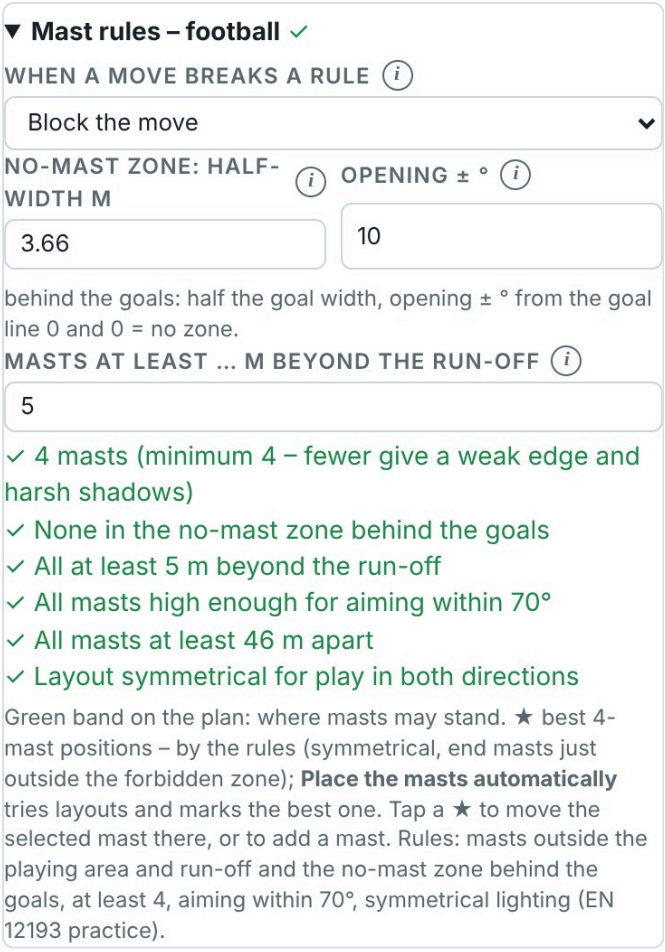
The demo floodlights are generic. A tender design needs the maker's photometry for the exact optic – beam angle alone does not describe a floodlight.

CHAPTER 7

Masts and mast rules



Masts. Number, luminaires per mast, place round the boundary, or take the masts from the stadium layout in Cable distribution.



Mast rules – football. Each rule with its status; set what happens when a move breaks one.

The rules are the sport's good practice made checkable: for football, at least 4 masts; none in the no-mast zone behind the goals (half the goal width plus an opening angle); at least 5 m beyond the run-off; high enough to aim within 70° of vertical; a minimum distance apart; and a layout symmetrical for play in both directions. **When a move breaks a rule** you choose: block the move, or allow it with a warning.

The **green band** on the plan shows where masts may stand; ★ **best locations** are the positions the tool found best for this ground and class. Tap a ★ to move the selected mast there.

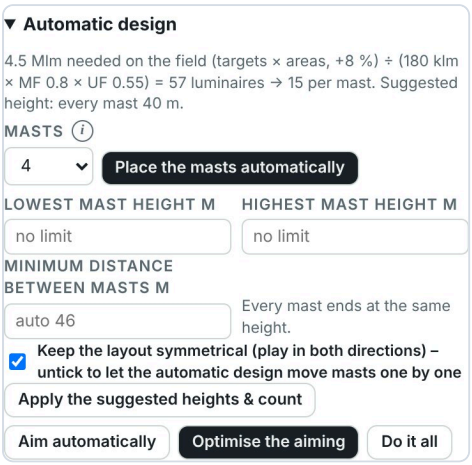
TIP – LINKED MASTS

When the stadium layout is placed in Cable distribution, tick **Masts follow the stadium layout**: the masts come here with their positions and heights and update whenever the electrical layout changes. **Send to the electrical design** sends the floodlights, kW and groups back for the cables.

CHAPTER 8

The automatic design

One button – **Complete the design automatically** – does what takes hours by hand. Here it took 28 seconds.



Automatic design. The suggestion, mast count and the steps you can run one at a time.

What it does, in order

1. **Places the masts** (if locations may change) – tries layouts within the rules and keeps the best.
2. **Sets heights and floodlight numbers** from the 25° rule, the 70° aiming limit and the lumen method.
3. **Aims and optimises** every floodlight, moving one aim point at a time while the shortfall against every target falls – up to four rounds, adding floodlights where still short.
4. **Trims a passing design** – lowers the masts towards the lowest allowed, then removes floodlights, each step kept only while every check still passes.
5. **Repeats for each ticked luminaire** and the mix, and keeps the best.
6. **With scenes**, sets each scene's level to the lowest that passes.

The message says what it did – *Design complete – every target met. 40 floodlights, masts 32 m (lowest allowed 32 m), 60.0 kW* – or, if it cannot pass, **why**: masts too low, too few, too far, or a class beyond the luminaire.

Running the steps yourself

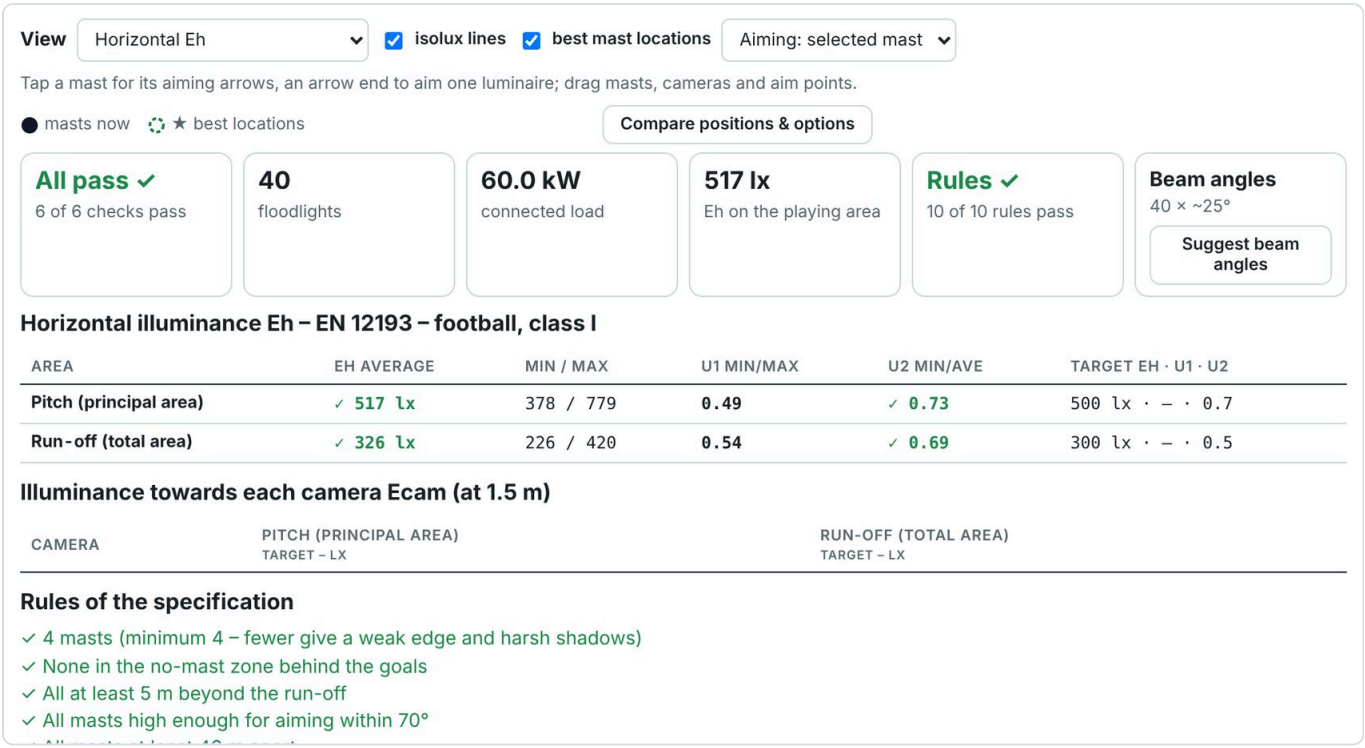
Place the masts automatically, **Apply the suggested heights & count**, **Aim automatically**, **Optimise the aiming**, or **Do it all**. Useful when masts are fixed and you only want the aiming redone.

WATCH OUT

A must-pass broadcast scene drives the design: in the case study, requiring 1800 lx to the cameras took the stadium from 48 floodlights and 72 kW to 628 floodlights and 942 kW – and still short. Decide early whether a televised scene must pass or is reported only (chapter 12).

CHAPTER 9

Reading the results



Results after the design. Cards on top; the Eh table per area against the targets; Ecam per camera; the rules; glare per end.

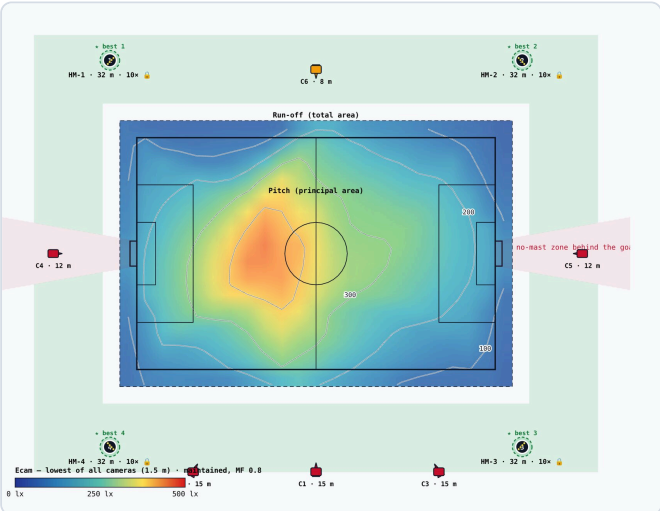
The values

- **Eh** – horizontal illuminance on the playing surface, maintained.
- **Ecam** – illuminance on a plane at 1.5 m facing each camera.
- **U1 = Emin / Emax**, **U2 = Emin / Eave** – uniformity.
- **GR** – glare rating (CIE 112) for players looking towards each mast.

✓ green passes, ✕ red fails, with the shortfall in the cell.

The views

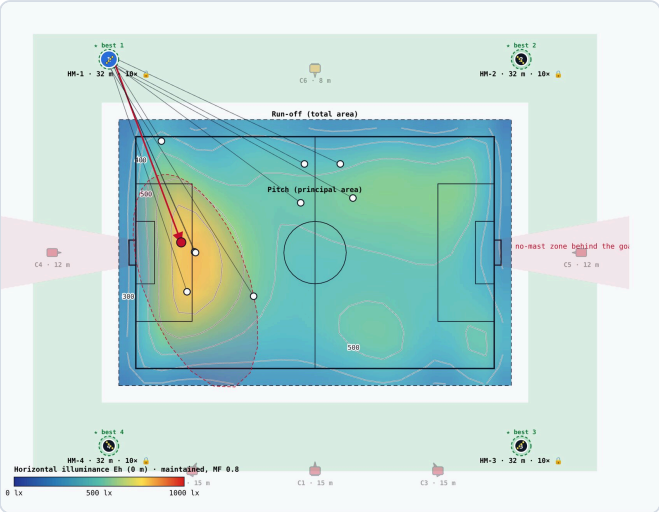
The View menu changes what the plan colours show: Eh, Ecam to any camera, the lowest Ecam of all cameras at each point (the hardest case for TV), and more. Isolux lines, the aiming arrows (none / the selected mast / all) and the best mast locations switch on and off.



Ecam – the lowest of all cameras at each point.

CHAPTER 10

Aiming by hand



One floodlight selected: its aim line in red and its 50 % beam footprint on the pitch.

1. **Tap a mast** – its aiming arrows appear.
2. **Tap an arrow end** to select one floodlight – its beam footprint shows.
3. **Drag the aim point.** The colours update as you drag (incremental calculation).
4. **Optimise** afterwards to tidy the rest around your change.

Every aim stays within 70° of vertical – a point dragged further is pulled back along its line.

When a mast moves

Each mast's **Aiming when moved** setting, shown by the symbol after its name: 🔒 aim points stay on the field; ⇄ aiming moves with the mast; ↻ re-aim automatically.

▼ **Heights & floodlights** 4 masts · 40 floodlights

HEIGHT OF ALL MASTS M ⓘ — 32

FLOODLIGHTS ON EVERY MAST ⓘ — 10

All masts +1 m | −1 m | All masts +2 floodlights | −2 | +10 %

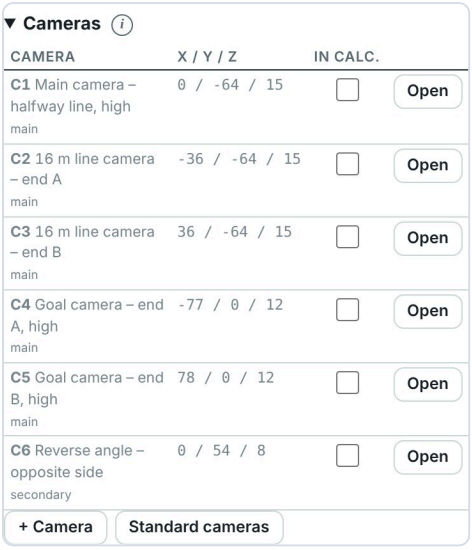
MAST	HEIGHT M	FLOODLIGHTS	FRAME	
HM-1	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-2	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-3	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op
HM-4	32	10	3.2 × 2.7 m 4 per row × 3 rows	Op

Changing a count keeps the aiming – new floodlights copy the existing aim points. Then optimise the aiming. ↶ Undo goes back one step.

Heights & floodlights. All masts at once (+1 m, +2 floodlights, +10 %) or each mast; its frame size and rows. New floodlights copy the existing aim points – optimise after.

CHAPTER 11

TV cameras

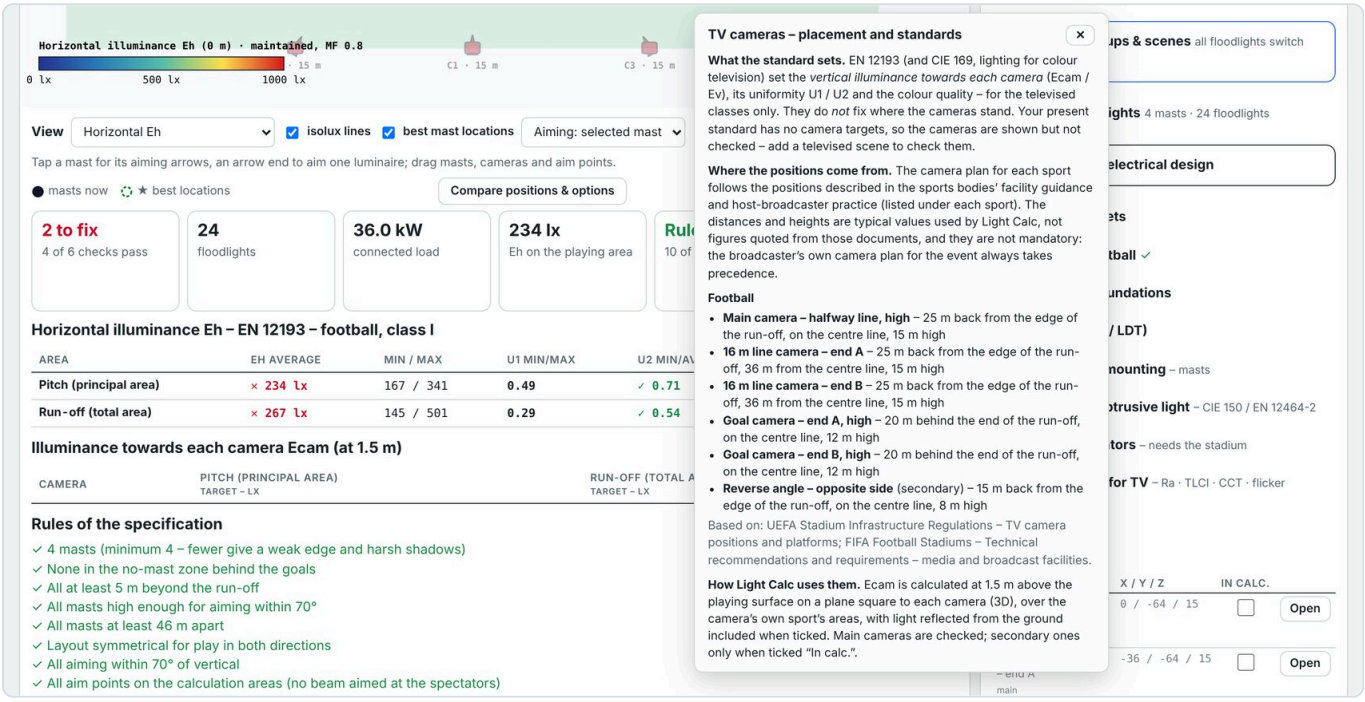


Cameras. Standard positions for the sport, each ticked in or out of the calculation.

Standard cameras places the usual camera plan for the sport: main camera on the halfway line, 16 m line cameras, goal cameras, reverse angle. Tick a camera **In calc.** and the Ecam values towards it appear in the results.

Drag a camera and it shows ⚠ **moved from the standard position**; ⟲ **Standard position** beside its Open button puts it back – one button puts back all moved cameras.

The **i** next to Cameras explains the placement, the standards behind it, where the positions come from, and how Light Calc uses them.



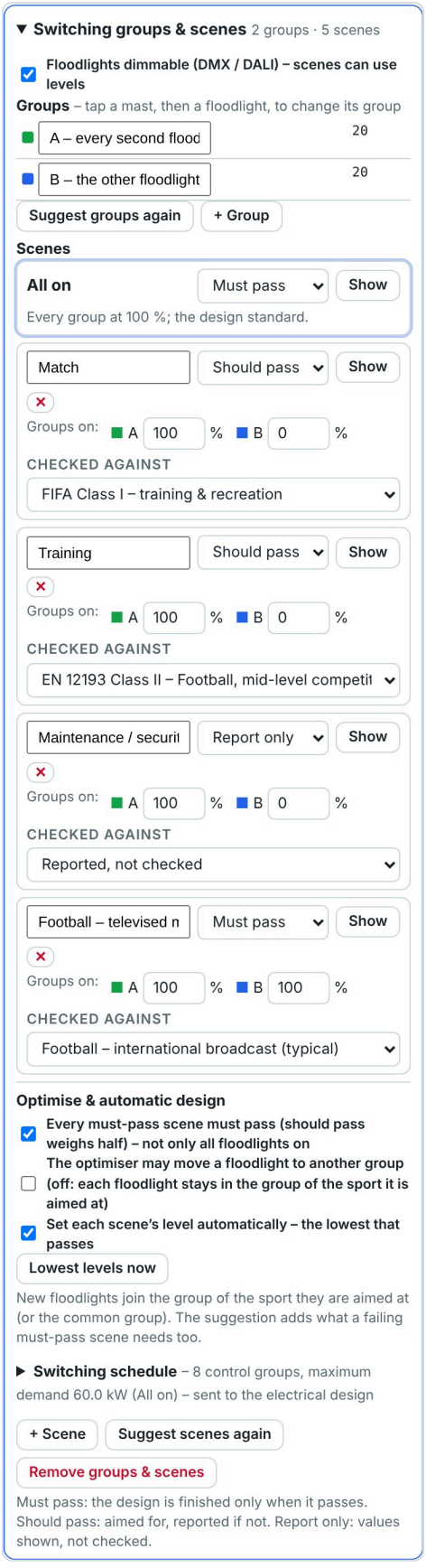
The camera i. What the standards ask, where the positions come from, how Ecam is calculated.

CAMERAS COUNT IN TELEVISED SCENES

A standard without camera targets (EN 12193 Class I–III) does not check the cameras. Light Calc points this out and offers two ways: a televised scene that must meet a broadcast class, or a report-only scene giving the broadcaster the camera values from the present lighting.

CHAPTER 12

Switching groups, scenes and DMX



Switching groups & scenes.

Groups

Floodlights are put into control groups – suggested as *A – every second floodlight* and *B – the other*, or by mast or sport. Tap a mast, then a floodlight, to change its group. Tick **dimmable (DMX / DALI)** and scenes can use levels; untick for contactors.

Scenes

Each scene sets every group's level and the class it is checked against, with a requirement:

- **Must pass** – the design is finished only when it passes.
- **Should pass** – aimed for, reported if not.
- **Report only** – values shown, not checked (security, present-lighting TV).

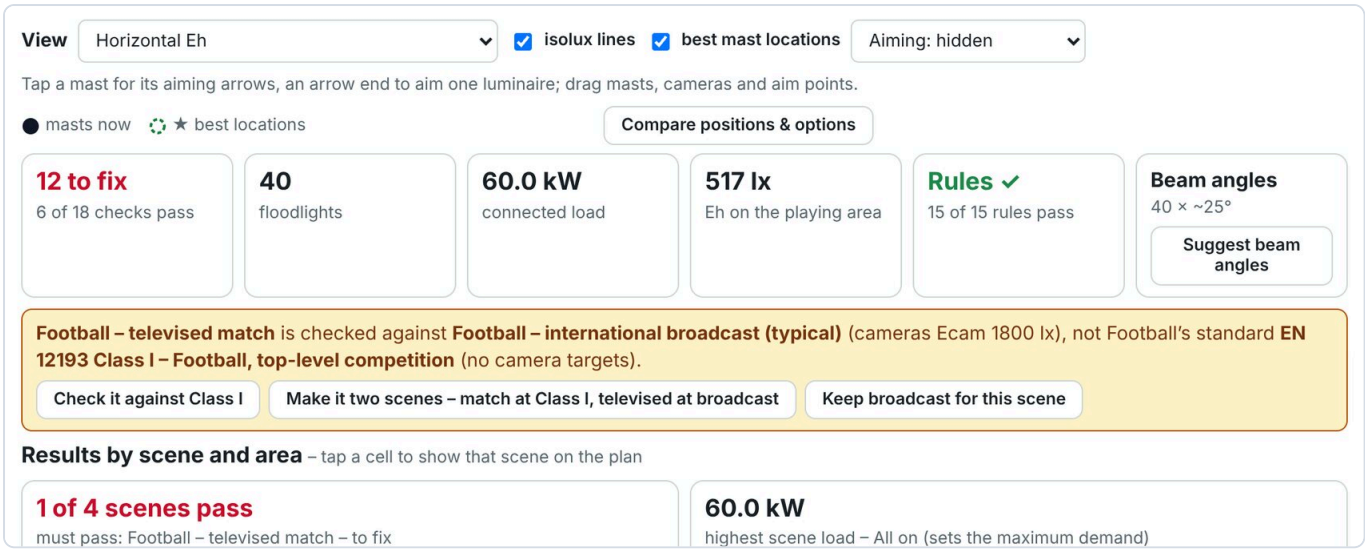
Suggest scenes creates the usual set for the sport: match, training, televised, maintenance / security.

Optimise and automatic design with scenes

Options: every must-pass scene must pass; the optimiser may move a floodlight to another group; set each scene's level automatically to the lowest that passes.

Switching schedule

Control groups with their floodlights and kW, each scene's groups and levels, and the **maximum demand** – the highest scene load, which sizes the supply.



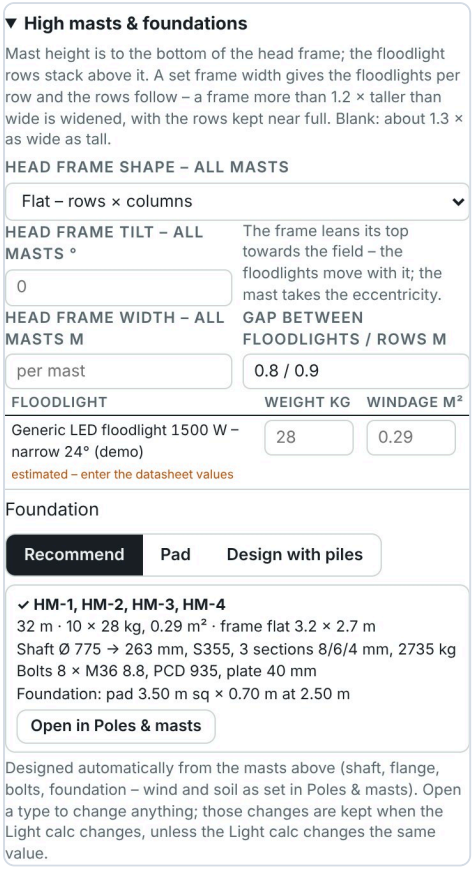
Results by scene. Scenes added after the design: *Match* and *Training* now switch only half the floodlights and fail. The yellow banner spots that the televised scene is checked against a broadcast class, not the club standard, and offers three ways out. Pressing **Complete the design automatically** again designs every scene together.

TIP – FOLLOW THE STANDARD

Change the sport's standard and the scenes that used it follow. A must-pass scene set to a higher class than the standard is flagged with the way back.

CHAPTER 13

High masts and foundations



High masts & foundations.

Each mast type gets a structural design from its height and head load:

- **Head frame** – shape (flat, stepped, curved, V, trapezoid, ring), tilt, width and gaps; floodlights per row and rows follow.
- **Floodlight weight and windage** – from the maker, or estimated from the wattage (marked as estimated).
- **Shaft** – diameter at the base and top, steel grade, sections and thicknesses, mass.
- **Base** – bolts, PCD, plate.
- **Foundation** – recommended, pad, or piles, with sizes.

The wind and soil come from Poles & masts; **Open in Poles & masts** shows the full calculation. Changes made there are kept for that type.

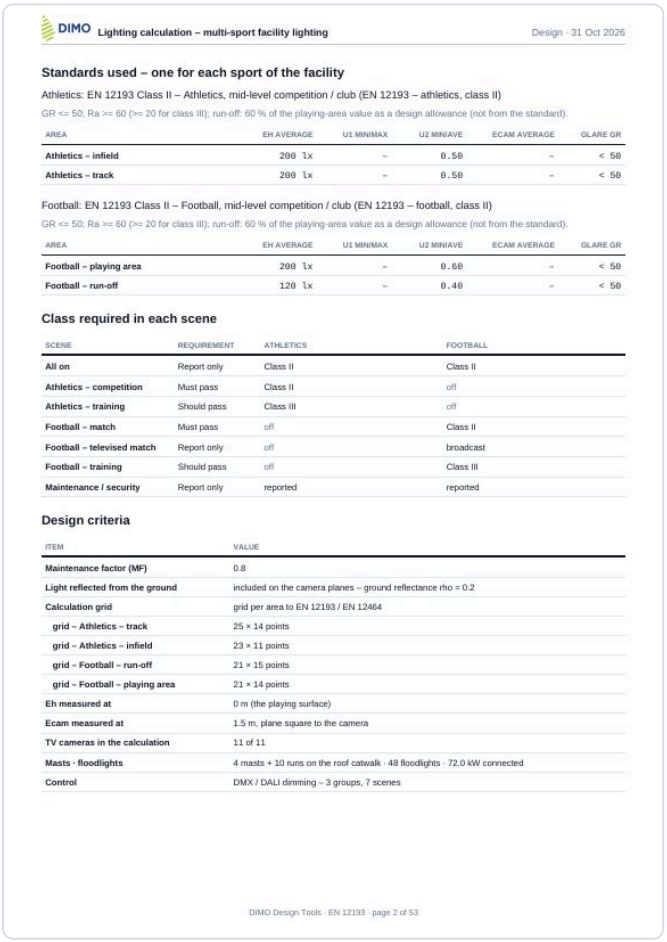
WATCH OUT

Enter the maker’s weight and windage. Estimates are marked in red until you do, and the foundation follows them.

CHAPTER 14

Reports and exports

BUTTON	WHAT YOU GET
PDF report	Summary cards and plan; the standards used (every sport's in a multi-sport design) and design criteria; Eh, Ecam, U1/U2 and glare tables; every scene with its plan and results; obtrusive light, stands, colour & flicker when used; luminaires; masts and roof sections, each head frame drawn with every floodlight numbered and coloured by group and its level in each scene; DMX control groups.
Aiming PDF	One section per mast: its aiming plan and every floodlight's aim point, tilt, direction and distance – for the installer.
Point values PDF	Every calculation point's values – for an approving engineer.
Page size	Auto, A3, A2 or A1 for the plans.
Import Relux .rdf	Opens a Relux project – floodlights as built, with their photometry and aiming.
Export new .rdf	A new Relux project with every floodlight, aiming, photometry, the ground and the main camera – no Relux file of your own needed.
Export plan .dxf	Ground lines, areas, masts, aiming lines, cameras, stands, roof and catwalk on their own layers, at the export position.
Send to electrical	Masts, floodlights, kW and control groups to the electrical design.



DIMO Lighting calculation – multi-sport facility lighting

Design · 31 Oct 2026

Standards used – one for each sport of the facility

Athletics: EN 12193 Class II – Athletics, mid-level competition / club (EN 12193 – athletics, class II)

GR <= 50; Ra >= 60 (>= 20 for class III); run-off: 60 % of the playing-area value as a design allowance (not from the standard).

AREA	EH AVERAGE	U1 MIN/MAX	U2 MIN/AVE	ECAM AVERAGE	GLARE GR
Athletics – infield	200 lx	–	0.50	–	< 50
Athletics – track	200 lx	–	0.50	–	< 50

Football: EN 12193 Class II – Football, mid-level competition / club (EN 12193 – football, class II)

GR <= 50; Ra >= 60 (>= 20 for class III); run-off: 60 % of the playing-area value as a design allowance (not from the standard).

AREA	EH AVERAGE	U1 MIN/MAX	U2 MIN/AVE	ECAM AVERAGE	GLARE GR
Football – playing area	200 lx	–	0.60	–	< 50
Football – run-off	120 lx	–	0.40	–	< 50

Class required in each scene

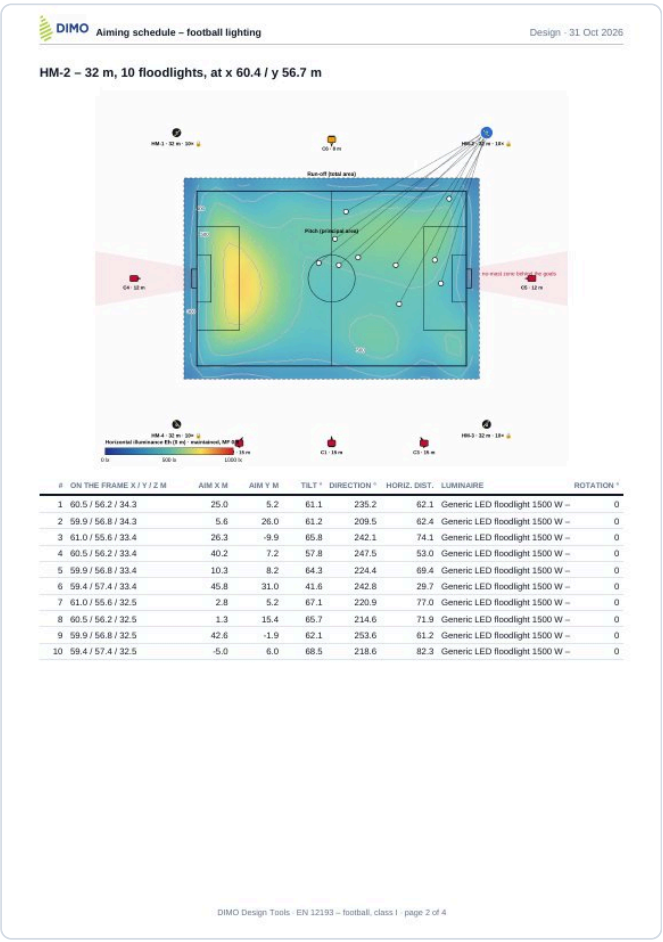
SCENE	REQUIREMENT	ATHLETICS	FOOTBALL
All on	Report only	Class II	Class II
Athletics – competition	Must pass	Class II	off
Athletics – training	Should pass	Class III	off
Football – match	Must pass	off	Class II
Football – televised match	Report only	off	broadcast
Football – training	Should pass	off	Class III
Maintenance / security	Report only	reported	reported

Design criteria

ITEM	VALUE
Maintenance factor (MF)	0.8
Light reflected from the ground	included on the camera planes – ground reflectance rho = 0.2
Calculation grid	grid per area to EN 12193 / EN 12464
grid – Athletics – track	25 × 14 points
grid – Athletics – infield	23 × 11 points
grid – Football – run-off	21 × 15 points
grid – Football – playing area	21 × 14 points
Eh measured at	0 m (the playing surface)
Ecam measured at	1.5 m, plane square to the camera
TV cameras in the calculation	11 of 11
Masts · floodlights	4 masts + 10 runs on the roof catwalk · 48 floodlights · 72.0 kW connected
Control	DMX / DALI dimming – 3 groups, 7 scenes

DIMO Design Tools · EN 12193 · page 2 of 53

Report: standards used, one per sport.



DIMO Aiming schedule – football lighting

Design · 31 Oct 2026

HM-2 – 32 m, 10 floodlights, at x 60.4 / y 56.7 m

#	ON THE FRAME X / Y / Z M	AIM X M	AIM Y M	TILT °	DIRECTION °	HORIZ. DIST.	LUMINAIRE	ROTATION °
1	60.5 / 56.2 / 34.3	25.0	5.2	61.1	235.2	62.1	Generic LED floodlight 1500 W –	0
2	59.9 / 56.8 / 34.3	5.6	26.0	61.2	209.5	62.4	Generic LED floodlight 1500 W –	0
3	61.0 / 55.6 / 33.4	26.3	-9.9	65.8	242.1	74.1	Generic LED floodlight 1500 W –	0
4	60.5 / 56.2 / 33.4	40.2	7.2	57.8	247.5	53.0	Generic LED floodlight 1500 W –	0
5	59.9 / 56.8 / 33.4	10.3	8.2	64.3	224.4	69.4	Generic LED floodlight 1500 W –	0
6	59.4 / 57.4 / 33.4	45.8	31.0	41.6	242.8	29.7	Generic LED floodlight 1500 W –	0
7	61.0 / 55.6 / 32.5	2.8	5.2	67.1	220.9	77.0	Generic LED floodlight 1500 W –	0
8	60.5 / 56.2 / 32.5	1.3	15.4	65.7	214.6	71.9	Generic LED floodlight 1500 W –	0
9	59.9 / 56.8 / 32.5	42.6	-1.9	62.1	253.6	61.2	Generic LED floodlight 1500 W –	0
10	59.4 / 57.4 / 32.5	-5.0	6.0	68.5	218.6	82.3	Generic LED floodlight 1500 W –	0

DIMO Design Tools · EN 12193 – football, class I · page 2 of 4

Aiming PDF: one mast's aiming plan and schedule.

PART C

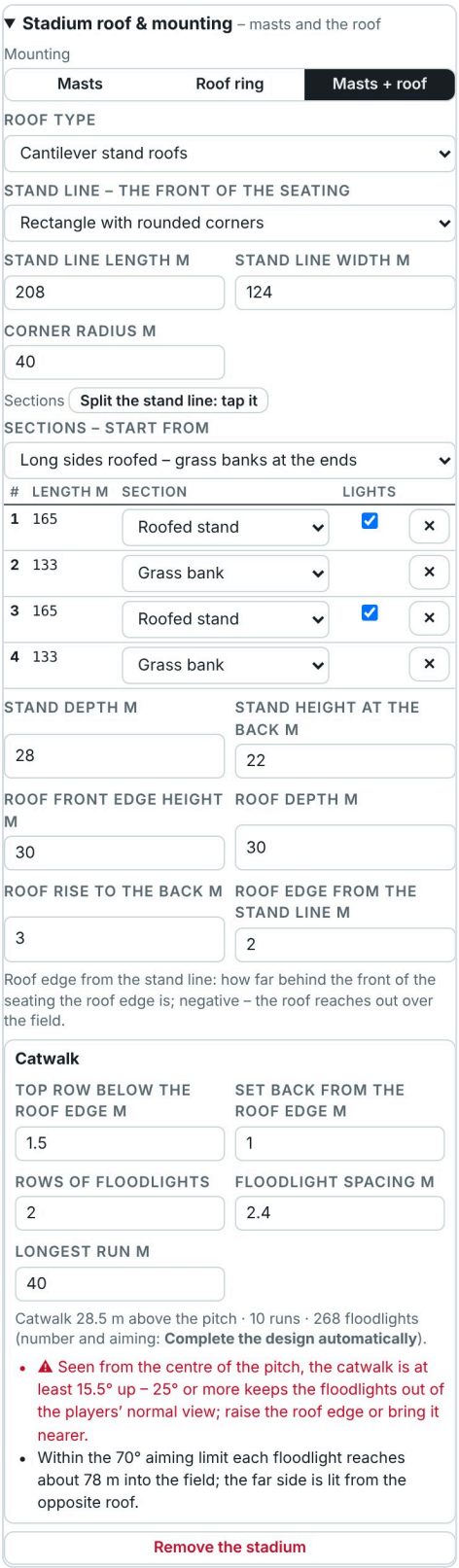
The powerful features

Roof-mounted floodlights, the photo-real view, the checks planning and broadcasters ask for, and how far you can trust the numbers.

CHAPTER 15

Stadium roof and roof mounting

Draw the stadium round the field – stands, grass banks, the roof – and mount floodlights on the roof catwalk, on masts, or both.



Stadium roof & mounting.

Mounting

Masts – floodlights on masts; the stadium is drawn and its roofs shade the masts' light. **Roof ring** – floodlights on the roof catwalk. **Masts + roof** – both. Your masts are kept when you switch.

Roof type

Cantilever, column-supported canopy, continuous ring (bowl), cable-net, arch-supported, retractable, dome, segmented curved cantilever. Ring, cable-net, retractable and dome roofs are carried all round, so every section is roofed.

Stand line and sections

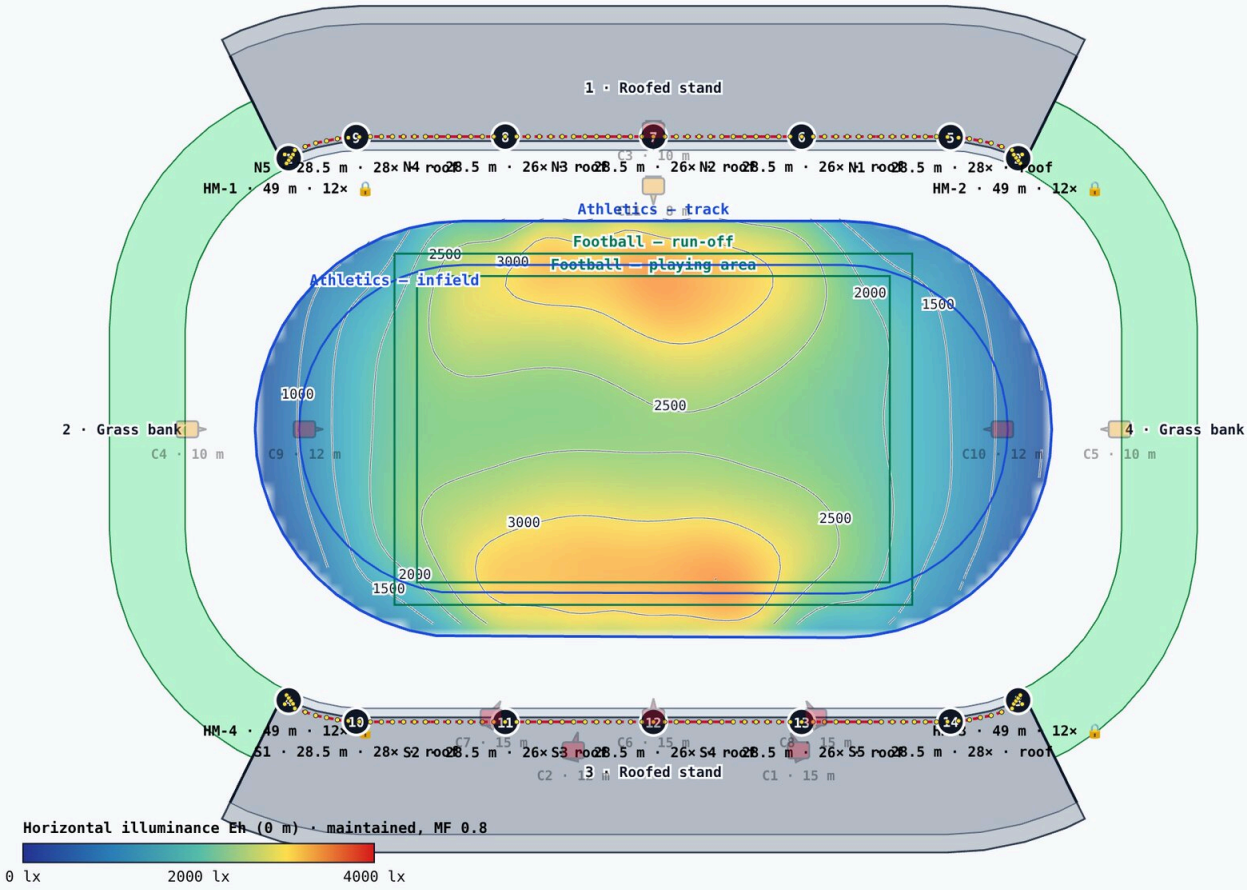
The **stand line** is the front of the seating: a rounded rectangle, an oval, or **traced on the plan** (tap round it on your drawing). It is cut into sections – roofed stand, open stand, grass bank or none – from a preset or by tapping the line. Each roofed section can carry floodlights or not.

Sizes and the catwalk

Stand depth and height; roof front-edge height, depth, rise and edge position. Catwalk: top row below the roof edge, set back, rows (1–3), spacing, longest run.

Runs work like masts

The catwalk is cut into runs (N1, N2 … S1 …). Each run is designed, aimed, optimised, grouped and reported exactly like a mast; its height stays the catwalk height.



On the plan: stands (grey), grass banks (green), roofs, the catwalk runs (red dashes) and four corner masts – Masts + roof.

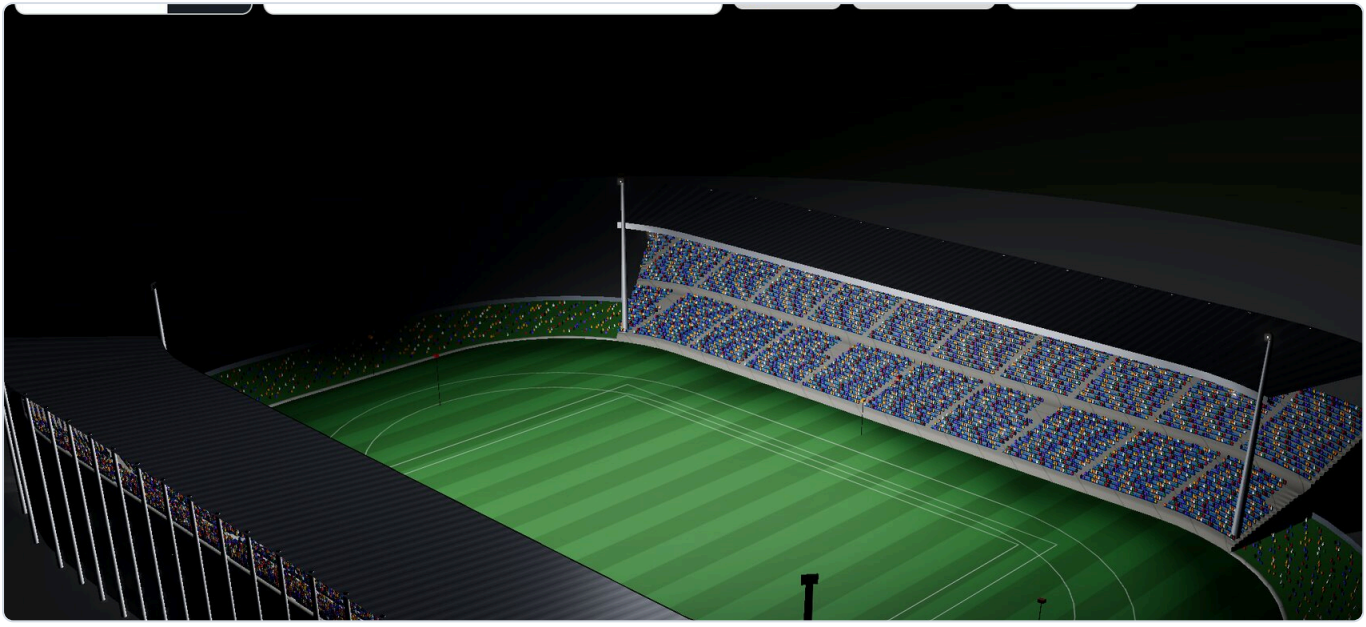
CHECKS FOR ROOF MOUNTING

The 25° line from the centre of the pitch to the catwalk (lower floodlights glare into the players' view), and how far each floodlight reaches within the 70° aiming limit – the opposite roof lights the far side.

CHAPTER 16

The realistic view

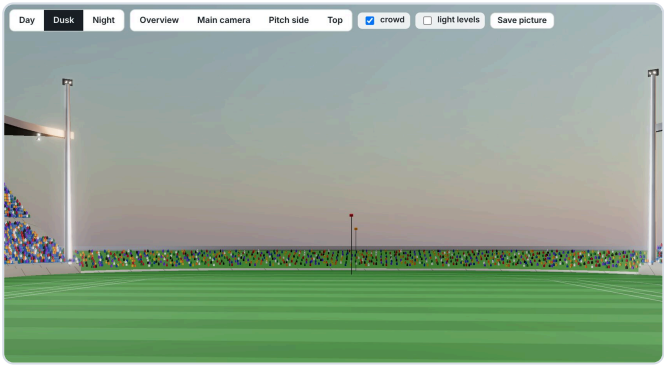
Plan · 3D · **Realistic**: the stadium, masts and floodlights as they will look – sky and sun, materials, seats and crowd, the roof by type, every floodlight aimed as designed.



Night. The pitch brightness comes from the calculated illuminance – a club-level pitch looks dimmer than a broadcast one. The caption gives floodlights, average lux and seats.



Day – for the architectural picture.



Dusk, pitch side.

Controls: Day · Dusk · Night; Overview, Main camera, Pitch side, Top; crowd on or off; light-levels overlay; **Save picture** (PNG for proposals). Drag to turn, scroll or pinch to zoom, right-drag to move.

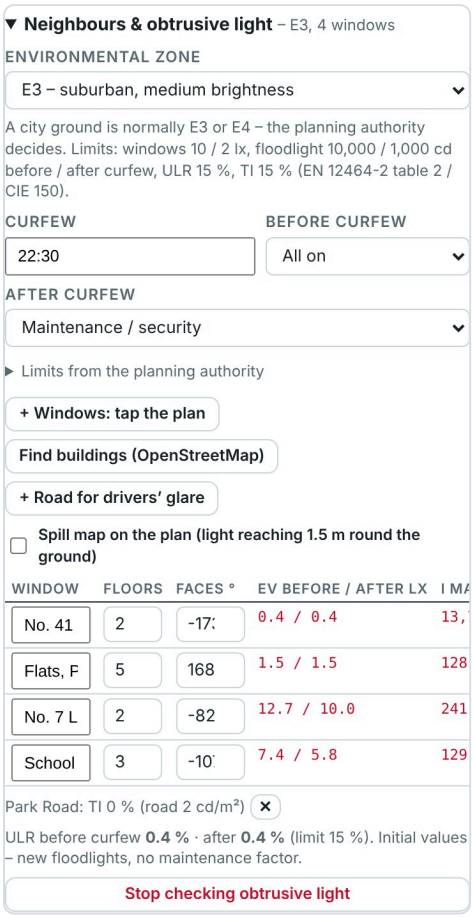
KEEP IN MIND

The view loads its 3D library the first time it opens, so it needs an internet connection then; Plan and 3D work offline. Only the pitch brightness is calculated – light on the stands and roof is for looks. Turn the crowd off on older phones.

CHAPTER 17

Neighbours and obtrusive light

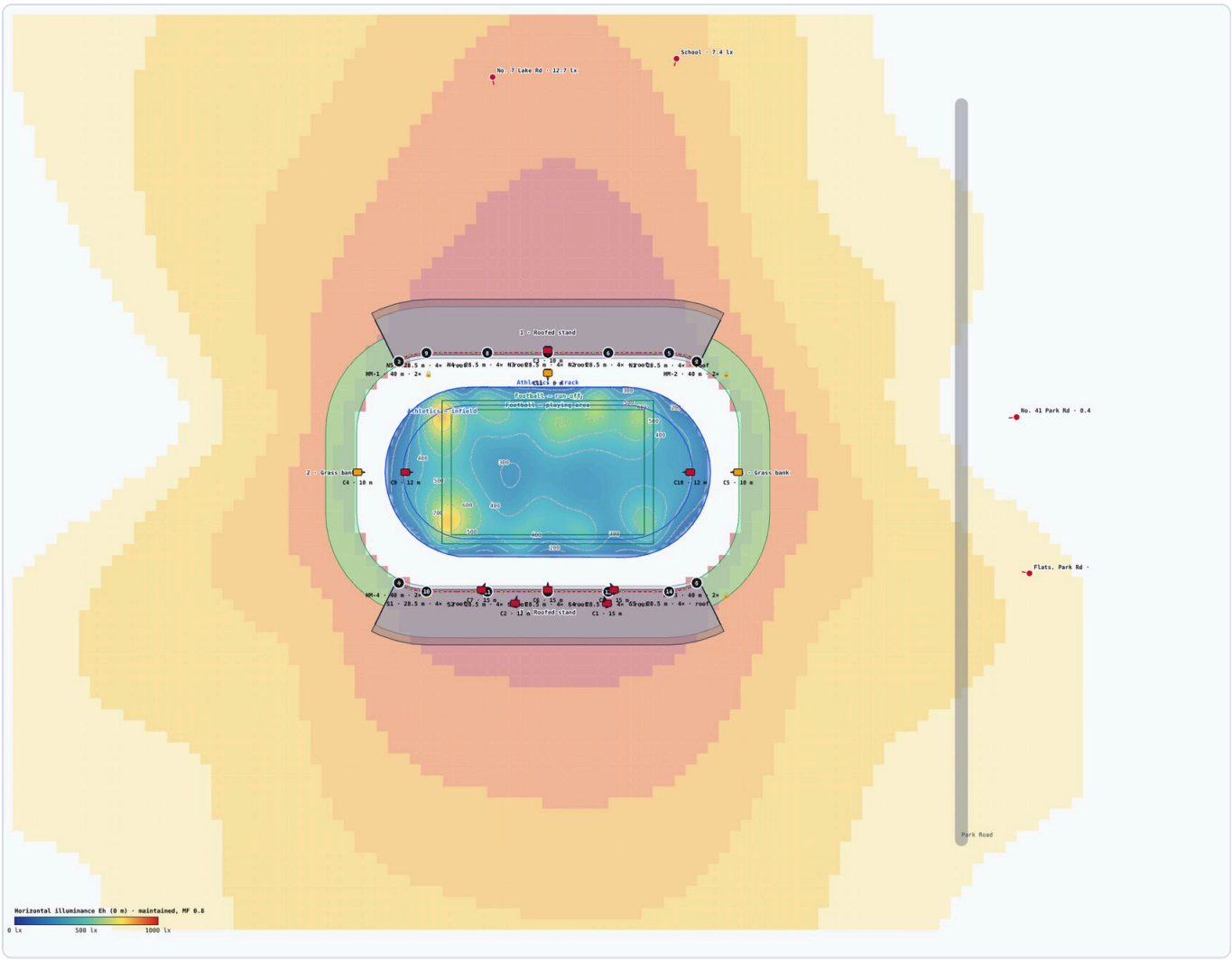
The check planning approval asks for near housing: light on neighbours' windows, the brightest floodlight they see, light going up into the sky, and glare for drivers – before and after the curfew, against the limits of the area's environmental zone (CIE 150 / EN 12464-2).



Neighbours & obtrusive light.

1. **Check obtrusive light**, then the **environmental zone** – E1 natural … E4 city centre. The planning authority decides; a city ground is usually E3 or E4. Its own figures can be typed instead.
2. **Curfew** and the scenes **before** and **after** it (usually match before; security or all off after).
3. **Windows** – tap the plan, or **Find buildings (OpenStreetMap)**: the 30 nearest buildings each get a window on the façade facing the ground, floors from the map. Set floors and direction in the table.
4. **Road** – tap along a road for drivers' glare (TI), both directions.
5. Optional **spill map** – light reaching 1.5 m round the ground.

Each window is calculated floor by floor like a camera plane – every floodlight, photometry and aim, with roofs blocking. Values are **initial** (new floodlights) – the worst case.



Spill map and windows on the plan – each window coloured pass / close / over, with its lux.

FIXING A FAILING WINDOW

Aim the floodlights that face it a few degrees lower; fit the maker's rear visor (load the visor's photometry); use an asymmetric optic; or lower the after-curfew scene. Each change shows on the window straight away.

CHAPTER 18

Stands and spectators

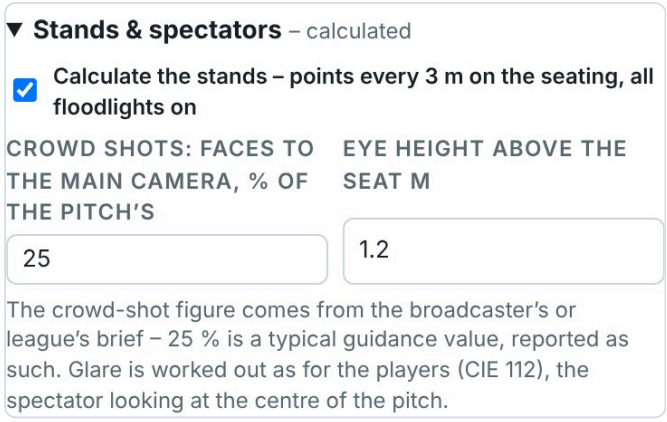
With the stadium drawn, tick **Calculate the stands**: points every 3 m on the seating, all floodlights on. For each section:

- **Eh on the seats**, average, minimum and U1.
- **Faces to the main camera** – vertical illuminance at eye height towards the camera, as a share of the pitch's. Stands behind the camera are marked *back to it*.
- **GR** – glare for a spectator looking at the centre of the pitch (CIE 112).

The crowd-shot share (default 25 %) comes from the broadcaster's or league's brief – it is guidance, not a standard requirement, and the report says so.

Roofs cutting the masts' beams

With masts behind roofed stands, each floodlight's 50 % beam is traced against the roof planes. Those losing 5 % or more are listed, with the mast height that clears them – or *re-aim* when none within 25 m does.



Stands & spectators.


DIMO Lighting calculation – multi-sport facility lighting

Design · 31 Oct 2026

Stands and spectators – all floodlights on

Calculation points every 3 m on the seating of each stand section. Eh: on the seats. Faces: vertical illuminance at 1.2 m above the seat towards Athletics – Main camera – home straight, finish area, high, against 25 % of the pitch's 334 lx (typical broadcaster guidance – not a standard requirement). GR: glare rating (CIE 112) for a spectator looking at the centre of the pitch.

SECTION	EH AV · MIN LX	U1	FACES LX	OF PITCH	GR MAX
1 Roofed stand · 165 m	6 · 0	0.01	32	✕ 10 %	✓ 36
2 Grass bank · 133 m	7 · 3	0.12	15	✕ 4 %	✓ 24
3 Roofed stand · 165 m	4 · 0	0.03	4	✕ 1 %	✓ 31
4 Grass bank · 133 m	9 · 4	0.12	18	✕ 5 %	✓ 18

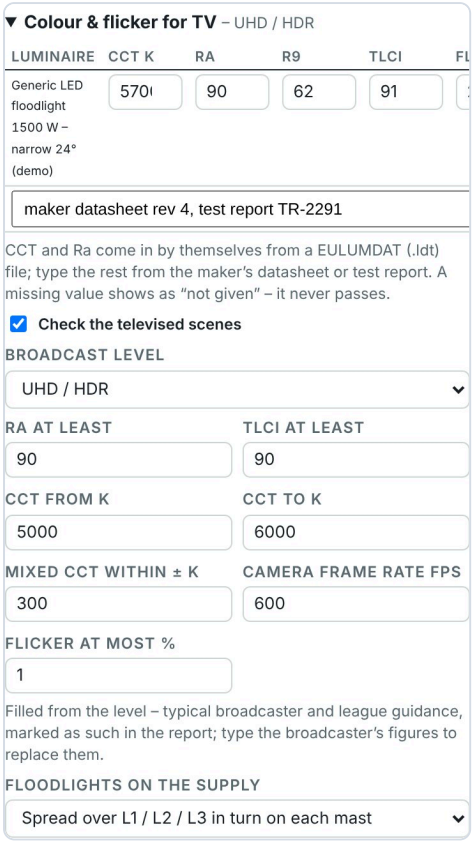
No mast floodlight loses 5 % or more of its beam to a roof.

DIMO Design Tools · EN 12193 · page 22 of 53

Report page – stands, with the case study's values.

CHAPTER 19

Colour and flicker for TV



Colour & flicker for TV.

Per luminaire

CCT, Ra, R9, TLCI, flicker % and the camera frame rate it is flicker-free to, with the source (datasheet, test report). CCT and Ra come in from LDT files. A missing value shows as *not given* – it never passes.

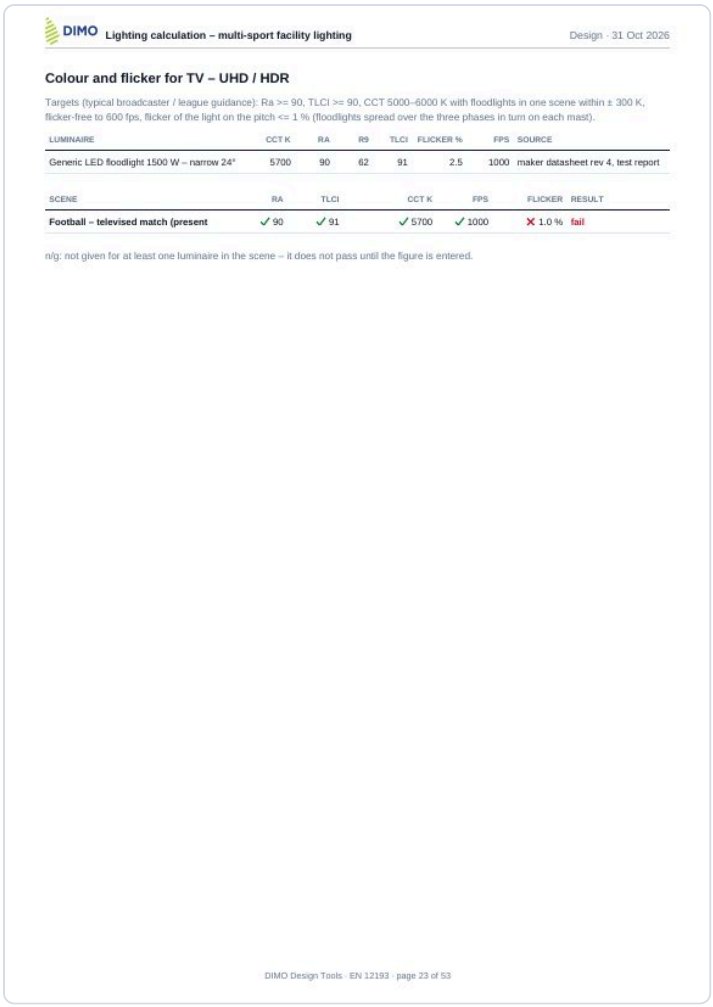
Per televised scene

Pick the broadcast level – **HD**, **UHD / HDR**, **super slow-motion** – and its targets fill in (editable; typical broadcaster and league guidance, marked as such in the report):

LEVEL	RA	TLCI	CCT K	± K	FPS	FLICKER
HD	80	80	4000–6500	500	300	3 %
UHD / HDR	90	90	5000–6000	300	600	1 %
Super slow-motion	90	90	5000–6000	300	1000	1 %

Combined flicker on the pitch

Floodlights on different phases ripple at different times and largely cancel. Light Calc adds each floodlight's ripple as a vector at each point of the pitch – floodlights spread over L1 / L2 / L3 in turn on each mast, or all on one phase as the worst case – and reports the highest.



DIMO Lighting calculation – multi-sport facility lighting

Design · 31 Oct 2026

Colour and flicker for TV – UHD / HDR

Targets (typical broadcaster / league guidance): Ra >= 90, TLCI >= 90, CCT 5000–6000 K with floodlights in one scene within ± 300 K, flicker-free to 600 fps, flicker of the light on the pitch <= 1 % (floodlights spread over the three phases in turn on each mast).

LUMINAIRE	CCT K	RA	R9	TLCI	FLICKER %	FPS	SOURCE
Generic LED floodlight 1500 W – narrow 24°	5700	90	62	91	2.5	1000	maker datasheet rev 4, test report

SCENE	RA	TLCI	CCT K	FPS	FLICKER	RESULT
Football – televised match (present	✓ 90	✓ 91	✓ 5700	✓ 1000	✗ 1.0 %	fail

n/g: not given for at least one luminaire in the scene – it does not pass until the figure is entered.

DIMO Design Tools · EN 12193 · page 23 of 53

Report page – the case study's televised scene: Ra and TLCI pass; the combined flicker of 1.01 % is just over the 1 % UHD target.

CHAPTER 20

Accuracy and validation

Every engine version is run against a suite of analytical test cases in the style of CIE 171:2006 – cases with exact answers worked out independently of the engine's code.

#	TEST	LIMIT
1–3	Point source: inverse-square and cosine laws; vertical planes; planes square to a TV camera	0.1 %
4–7	Aimed floodlights at 50–70° tilt; asymmetric distributions and spin; table interpolation; flux and IES = LDT	0.01–1 %
8	Four masts summed with the maintenance factor (full calculation)	1 %
9a/9b	Ground reflection onto camera planes – against fine numerical integration	5 / 10 %
10	Glare rating GR (CIE 112)	0.1 %
11/11b	Light blocked by stands – exact line-of-sight	0.01 %
12	Energy balance – flux reaching the ground = flux sent down	1 %
13	CIE 171:2006 case 5.6 – published reference values	5 %
14	Roof catwalk positions; roof shading a mast	0.5 %
15–17	Neighbour's window Ev and intensity; upward light ratio; drivers' TI	0.1–0.5 %
18	Combined flicker on three phases	0.01 %
19	No calculation points under a pavilion or closed structure; every point outside kept	exact

All 21 groups pass on v157. The calculation itself: point-by-point direct light from the IES / LDT tables (bilinear between table values), ground reflection on the camera planes with near-field rings and far-field patches, glare to CIE 112 within 1.5°–60°, and exact obstruction by structures and roofs.

INDEPENDENT CHECK

For a submission, export the design as a new Relux .rdf and run it in Relux – the grid follows the same standard spacing, so the tables compare directly.

CHAPTER 21

Twelve power tips

1. **Tick two or three luminaires for the automatic design.** It compares each and the mix, and Compare positions & options shows them side by side.
2. **Untick Dynamic plan on big designs** and press Calculate the changes when ready.
3. **Set the scenes before the automatic design** – it then designs every scene and sets each one's lowest passing level.
4. **Decide early: must-pass or report-only TV.** It can multiply the floodlights several times.
5. **Use ★ best locations** for the first layout, then let the rules guard your moves.
6. **Fixed masts?** Set Mast locations to Fixed and use Aim automatically + Optimise.
7. **Mix beams** with Suggest beam angles before adding floodlights.
8. **Mark structures** on the drawing – their shadows are exact.
9. **Link the masts to the stadium layout** and Send to electrical, so cables and lighting never drift apart.
10. **Run obtrusive light early** near housing – fixing aim and visors is cheap before the masts are ordered.
11. **Save a realistic picture** at night from the main camera for the proposal.
12. **Export a new .rdf** for the independent Relux check before submission.

PART D – CASE STUDY

A multi-sport stadium

Athletics and football on one ground, roofed long sides and grass banks at the ends, floodlights on corner masts and the roof catwalk, houses and a school nearby, and a broadcaster asking for camera values. Worked through in Light Calc from start to finish.

CHAPTER 22

The brief

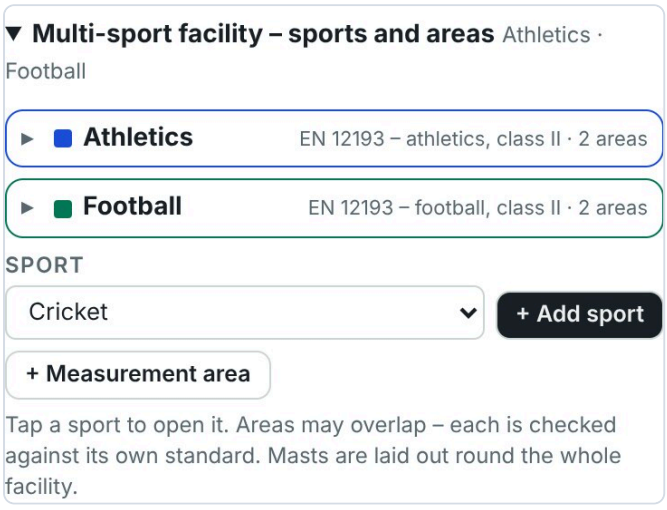
“Colombo City Sports Complex” – a fictional ground used for this example. The site, houses and luminaire data are invented; every result is Light Calc’s own.

ITEM	BRIEF
Sports	400 m athletics track, 8 lanes, with a football pitch inside
Standards	Athletics: EN 12193 Class II (club competition). Football: EN 12193 Class II
Scenes	Athletics competition (must pass), athletics training, football match (must pass), football training, televised football – camera values for the broadcaster (report only), maintenance / security
Stadium	Roofed stands on both long sides (cantilever roof, front edge 30 m); grass banks at both ends
Mounting	Masts + roof: four corner masts behind the grass banks, plus the roof catwalk over the long sides
Control	DMX dimming
Neighbours	Houses and flats on Park Road (east, about 255 m), houses and a school on Lake Road (north, about 215 m); zone E3; curfew 22:30
TV	UHD coverage of football; floodlight data: CCT 5700 K, Ra 90, R9 62, TLCI 91, flicker 2.5 %, flicker-free to 1000 fps (example values)

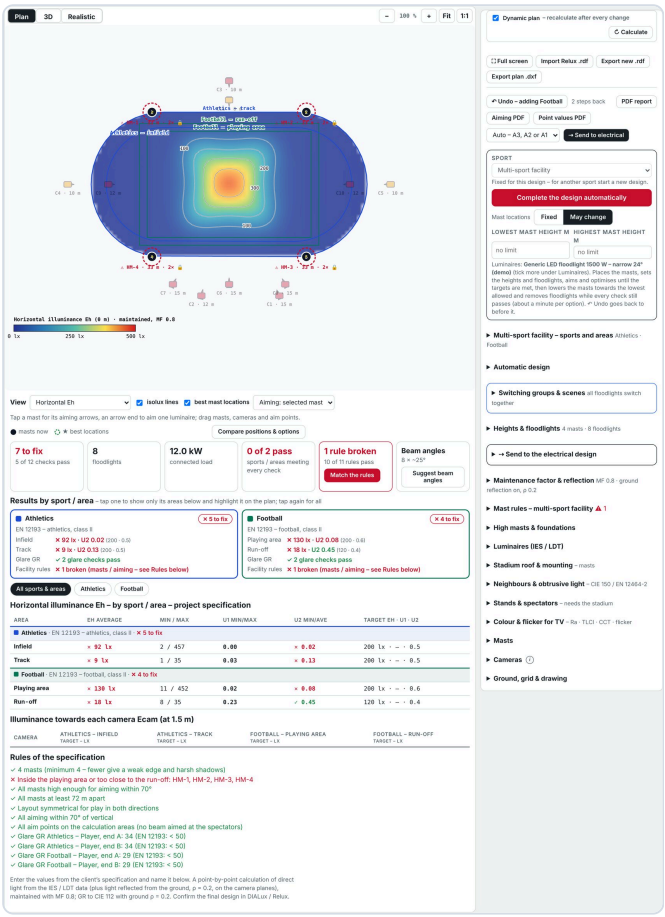
CHAPTER 23

Setting it up

1. **Sport: Multi-sport facility.** Add **Athletics**, then **Football**; centre the pitch on the track. Each sport gets its own areas – athletics infield and track, football playing area and run-off – and its own standard.
2. **Standards:** athletics EN 12193 Class II, football EN 12193 Class II, each set in its sport's line.
3. **Stadium:** stand line 208 × 124 m, corners 40 m; **Sections – Long sides roofed – grass banks at the ends**; cantilever roof, front edge 30 m, depth 30 m, stands 28 m deep.
4. **Masts:** four corner masts placed round the boundary; then **Mounting: Masts + roof**. The catwalk is cut into ten runs, N1–N5 and S1–S5, at 28.5 m.
5. **Scenes:** Suggest scenes, then **Report the cameras only – broadcast with the present lighting** for football; DMX ticked.



Sports and areas – each sport with its standard.



The ground – track, infield and pitch.

CHAPTER 24

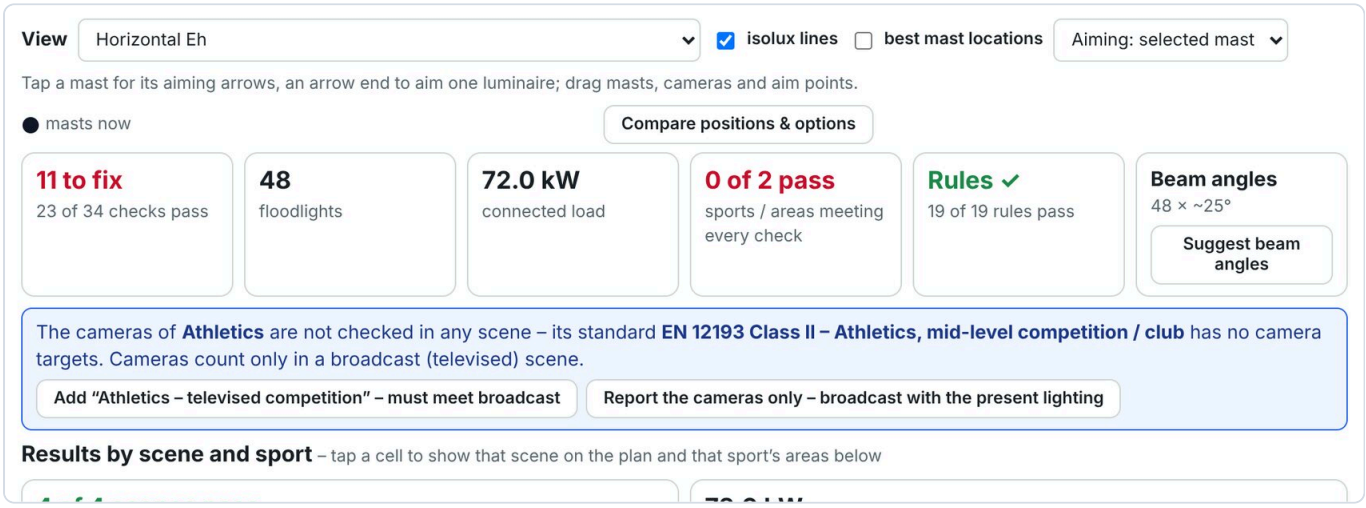
Designing and checking

Complete the design automatically – with the masts and catwalk fixed, it sized the floodlights on each, aimed and optimised them for every scene, and set each scene's level. 88 seconds.

48	72.0 kW	40 / 28.5 m	4 of 4
floodlights – 8 on the corner masts, 40 on the catwalk	connected load, all on	corner masts / catwalk	must and should scenes pass

SCENE	REQUIREMENT	LEVEL SET	KW	RESULT
Athletics – competition	must pass	65 %	46.8	pass
Athletics – training	should pass	35 %	25.2	pass
Football – match	must pass	50 %	36.0	pass
Football – training	should pass	20 %	14.4	pass
Football – televised (present lighting)	report only	100 %	72.0	camera values reported
Maintenance / security	report only	–	57.0	reported

Levels were set automatically to the lowest that passes: athletics competition 100 → 65 %, training 65 → 35 %, football match 85 → 50 %, training 35 → 20 %. All on, the playing area averages 433 lx and the track 314 lx.



Results. Light Calc notices that athletics' cameras are in no scene and offers a televised athletics scene or a report of the cameras. The top cards check All on against each sport's standard; with scenes, the scene table is what decides.

LESSON – THE BROADCAST TRAP

With the televised football scene set to **must** meet the international broadcast class (1800 lx to the cameras), the same run grew to 628 floodlights and 942 kW – and still missed by 20 lx. Reporting the cameras from the club design gave the broadcaster their values for 72 kW.

LESSON – THE SECURITY SCENE

The suggested security scene draws 57 kW – 79 % of the total. Set it to one group at 10–20 % before the obtrusive-light check: it is also the after-curfew scene.

CHAPTER 25

The neighbours, the stands and TV

Obtrusive light – zone E3, curfew 22:30

WINDOW	EV BEFORE	EV AFTER	I MAX	RESULT
No. 41 Park Road	0.4 lx	0.4 lx	13,792 cd	I over 10,000 / 1,000 cd
Flats, Park Road (5 floors)	1.5 lx	1.5 lx	128,267 cd	I over
No. 7 Lake Road	12.7 lx	10.0 lx	241,117 cd	Ev and I over
School, Lake Road	7.4 lx	5.8 lx	129,756 cd	Ev after and I over
Upward light ratio		0.4 %	limit 15 %	pass
Drivers' TI, Park Road		0.2 %	limit 15 %	pass

The light on the Park Road windows is low, but they look straight into catwalk floodlights aimed across the field. Lake Road and the school, across the field from the south catwalk, get both spill and intensity – and the after-curfew security scene keeps them over.

WHAT TO DO

(1) Make the security scene one group at 10–20 %, or all off after curfew. (2) Fit rear visors to the south-catwalk floodlights facing Lake Road and load the visor photometry. (3) Aim those floodlights 2–3° lower. Re-check: the windows update at once.

Stands – all on

SECTION	EH SEATS	FACES	SHARE	GR
North, roofed	6 lx	32 lx	10 %	36
West, grass bank	7 lx	15 lx	4 %	24
South, roofed	4 lx	4 lx	1 %	31
East, grass bank	9 lx	18 lx	5 %	18

Faces towards the main camera vs the pitch's 334 lx. Below the 25 % crowd-shot guidance everywhere – the design lights the field, not the crowd. A few floodlights aimed at the stands, or dedicated spectator lighting, would fix it. No mast floodlight loses 5 % of its beam to a roof.

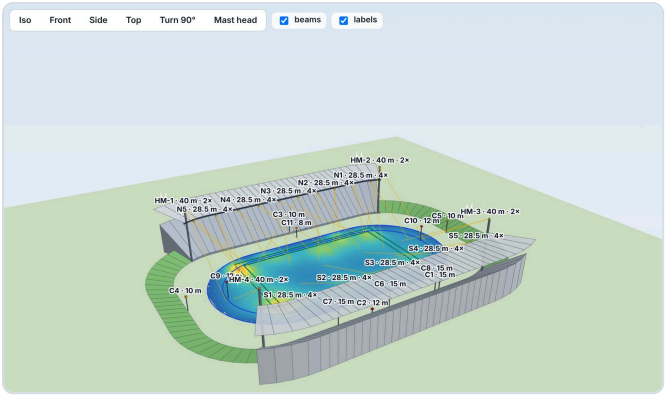
Colour and flicker – UHD

CHECK	VALUE	TARGET	
Ra	90	≥ 90	✓
TLCI	91	≥ 90	✓
CCT	5700 K	5000–6000	✓
Frame rate	1000 fps	≥ 600	✓
Flicker on the pitch	1.01 %	≤ 1 %	✗

2.5 % per floodlight, spread over three phases, leaves 1.01 % on the pitch – just over. On one phase it would be the full 2.5 %. A driver with lower flicker (≤ 2.4 %), or better phase balance on the catwalk, passes.

CHAPTER 26

Pictures, report and lessons

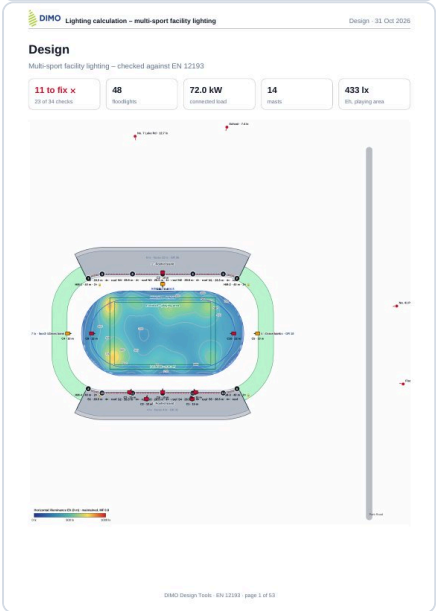


3D – masts, catwalk runs and beams with labels.



Realistic, day – for the client.

The report – 53 pages



Summary and plan



Obtrusive light



Roof section N1 – its floodlights along the catwalk

Lessons

1. **Set scenes first.** The automatic design then designs every scene and finds each one's lowest level – here saving up to 80 % in training.
2. **Decide what TV requires before designing.** Must-pass broadcast: 942 kW. Reported camera values: 72 kW.
3. **Check the neighbours early.** The security scene and two catwalk directions decide planning approval – both cheap to fix now, costly after the floodlights are bought.
4. **Light the stands on purpose** if crowd shots matter – spill alone gives 1–10 % of the pitch.
5. **Ask the maker for flicker and TLCI** early – a 0.01 % miss on flicker is a driver choice.

PART E – EVERY CONTROL

Drawings, maps and every editor

Three worked imports (a PDF site plan, an aerial picture or Google Maps screenshot, and OpenStreetMap), marking every kind of area, the ground fields for each sport, the multi-sport tabs and special areas, the mast and camera editors, the plan controls and pop-ups, and every “i” help text.

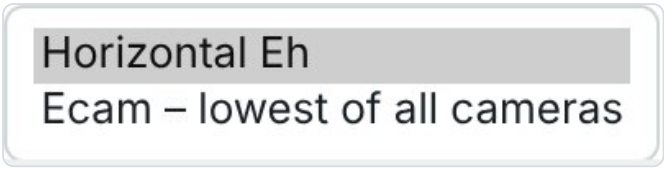
CHAPTER 27

Plan controls and pop-ups

Everything above, beside and under the plan, from left to right.



The three views. Plan (the calculation), 3D (masts, beams, roof) and Realistic (chapter 16).



View. The colour plan shows horizontal Eh, or Ecam – the lowest of all the cameras at each point.

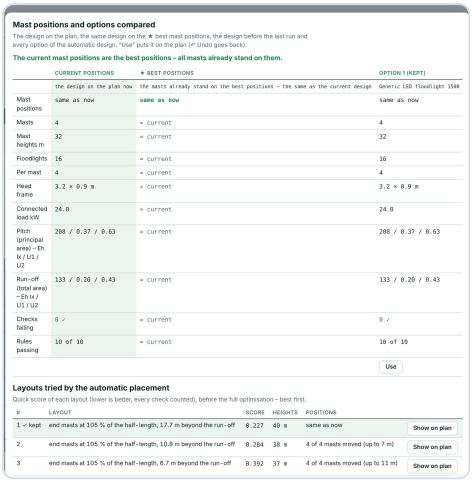
CONTROL	WHAT IT DOES
– / +	Zoom out and in.
Fit · 1:1	Fit the plan to the width of the screen · actual size (scroll to move round).
isolux lines	Draws the lines of equal lux (labelled every 100 lx) over the colour plan.
best mast locations	Shows the ★ best 1–4 positions found by the last placement (dashed circles). Tap a ★ to move the selected mast there, or to add a mast.
Aiming: selected mast / all masts / hidden	Aiming lines from the head frame to every aim point. Drag an aim point to re-aim one floodlight.
Dynamic plan	Recalculates after every change. Untick on large designs and press ↻ **Calculate** when ready.
⛶ **Full screen**	The plan fills the screen – best for marking on a drawing on a phone.
Iso · Front · Side · Top · Turn 90°	3D view only: the standard viewpoints, and turn the model a quarter at a time. Drag to turn it; the mouse wheel zooms.
Mast head	3D only: each press flies to the next mast's head frame, so you can check the rows, the tilt and the spacing.
beams · labels	3D only: the floodlight beams as cones; mast and camera names.
Point values PDF paper	Auto picks A3, A2 or A1 so every value stays readable; A1–A2 full size; A3 and A4 scaled to fit.

Compare positions & options

Opens a table side by side: the design on the plan now, the same design on the ★ best mast positions, and every option the last automatic design tried (each luminaire and the mix). Each column lists masts, heights, floodlights, head frame, connected load, the results of every area, the failing checks and the rules passing. **Use** puts that column on the plan; ↶ Undo goes back.

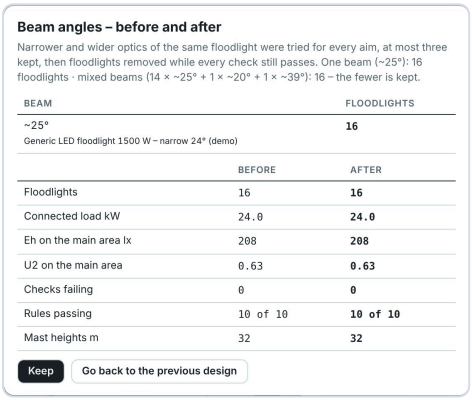
Below it, **Layouts tried by the automatic placement** scores every layout searched (lower is better, every check counted). **Show on plan** previews one.

> **TIP**
>
> When the client asks “why these positions?”, this table is the answer: the kept layout and the next five, with their scores.



Compare. Here the masts already stand on the best positions.

Suggest beam angles



Beam angles – before and after. Keep it, or go back.

For every aim, the narrower and wider optics of the same floodlight family are tried (at most three kept); floodlights are then removed while every check still passes. The pop-up compares floodlights, connected load, Eh and U2 on the main area, failing checks, rules passing and mast heights before and after. **Keep** applies it; **Go back to the previous design** undoes it.

> **WHILE IT RUNS**
>
> The design is locked during any automatic run – beam angles, the automatic design, placement or optimising. Views, zoom, 3D, the panes and help still work; anything else shows a short note. Press ■ **Stop** in Automatic design to end the run early.

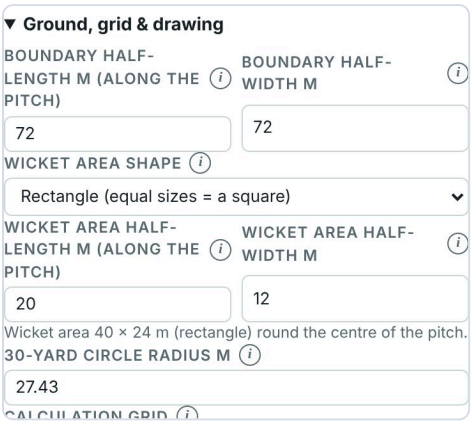
Match the rules

Appears in red when a mast rule is broken (for example the layout is not symmetrical). It moves, adds or raises masts until every mast rule holds, then improves the design until every check passes again. A pop-up compares the design before and after.

CHAPTER 28

The ground for each sport

The first fields under **Ground, grid & drawing** change with the sport. Each has an “i” with the rule behind it.

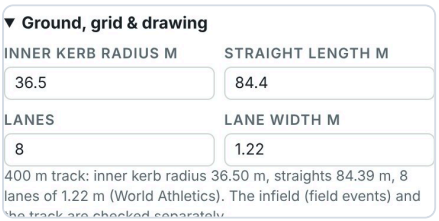


Cricket (Philips LiDAC).

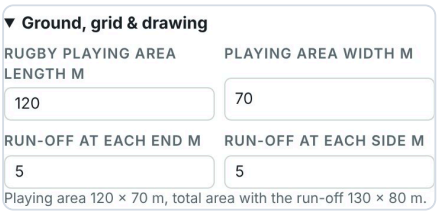
Cricket

FIELD	MEANING
Boundary half-length / half-width	From the centre of the pitch to the rope, along and across the pitch (59–82 m international). Equal for a round ground.
Wicket area shape	Rectangle (equal sizes = a square), the usual calculation area; or ellipse, as in the LiDAC sketch.
Wicket area half-length / half-width	The zone round the pitch needing the highest, most even light. 20 × 12 gives a 40 × 24 m area.
30-yard circle radius	27.43 m – two half-circles round the middle stumps joined by straight lines: the inner field.
Forbidden zone ± °	(Mast rules) No mast in line with the pitch behind the wickets, from the wicket-area edges.
Masts at least … m beyond the boundary	(Mast rules) Room for the boundary run-off, boards and sight screens.

Calculation areas: wicket area, inner field, outer field & boundary. Special area: **the strip**.



Athletics. Inner kerb radius 36.5 m, straights 84.39 m, 8 lanes of 1.22 m (the World Athletics 400 m track). Areas: infield (field events) and track (lanes).



Rugby. Playing area 120 × 70 m including in-goals, run-off 5 m. Special areas: in-goal areas, 22 m areas.



Tennis. 23.78 × 10.98 m, run-off 6.4 m at the ends, 3.66 m at the sides. Masts are called **poles**.

Football: playing area 105 × 68 m, run-off at each end and side. Special areas: penalty areas, goal areas, centre circle. **Pickleball and volleyball** work like tennis. Under the size the tool states the result, for example “Playing area 105 × 68 m, total area with the run-off 115 × 78 m”, and the grid it will use for each area.

ATHLETICS SPECIAL AREAS

Finish line (photo-finish camera) · sprint straight (100 m / 110 m hurdles) · long / triple jump runway and pit · pole vault runway and landing · high jump fan · shot put circle and sector · discus / hammer circle and sector · javelin runway and sector. Each is checked against its own target (chapter 33).

CHAPTER 29

Worked case – your PDF drawing

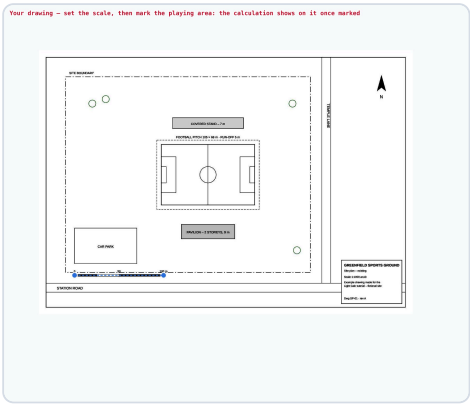
A club ground drawn by the architect: *Greenfield Sports Ground, site plan SP-01*, an A3 PDF at 1:1000 with a 0–100 m scale bar, a pavilion, a covered stand and the site boundary. Target: football EN 12193 Class II, masts only where the site allows.

1 · Load the drawing

Open **Ground, grid & drawing** and press **Choose File** under *Your drawing (image or PDF) under the plan*. A PDF's first page is turned into a picture at 2000 px across; an image (PNG, JPG) is used as it is.

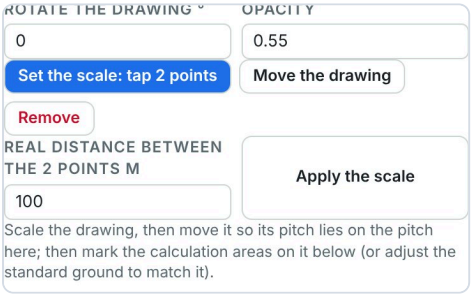
The red line over the plan says what comes next: *Your drawing – set the scale, then mark the playing area*. Until areas are marked, the standard ground is drawn over the drawing only if **Show the standard ground over the drawing** is ticked.

OPTION	USE
Show the drawing under the plan	Untick once traced – the drawing stays with the design.
Darken faint lines	Makes thin CAD lines readable.
Rotate the drawing °	Turn it so the pitch runs along x.
Opacity	0.55 is a good start.
Remove	Takes the drawing away.



Loaded. The site plan lies under the plan, not yet to scale.

2 · Set the scale



Set the scale: tap 2 points.

1. Press **Set the scale: tap 2 points** (it turns blue).
2. Tap the 0 and the 100 m end of the scale bar. Zoom in (+) first: the closer you tap, the better the scale.
3. Type the real distance, **100**, under *Real distance between the 2 points m*.
4. Press **Apply the scale**.

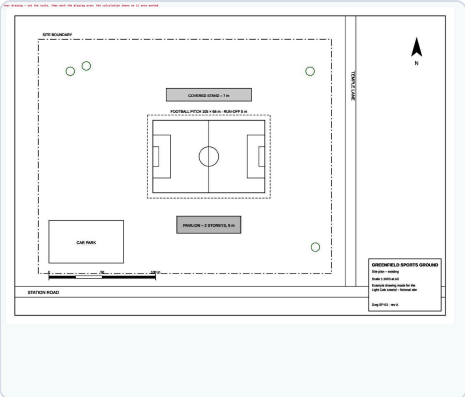
No scale bar? Use any known length: the pitch (105 m), a dimensioned building, or the grid of a survey.

CHECK

After scaling, the drawing's pitch should match the standard pitch outline in size. If it is a little off, scale again over a longer distance.

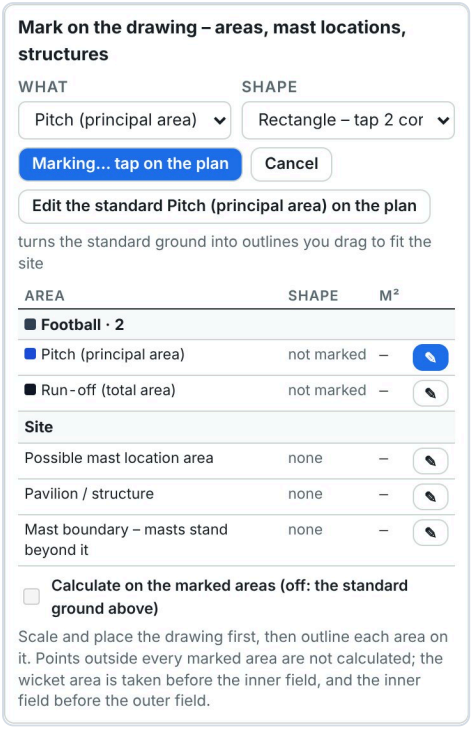
3 · Move the drawing

Press **Move the drawing** and drag the drawing until its pitch lies on the plan's pitch (the plan's 0, 0 is the centre spot). Press it again to stop.



Moved. The architect's pitch now lies on the calculation pitch.

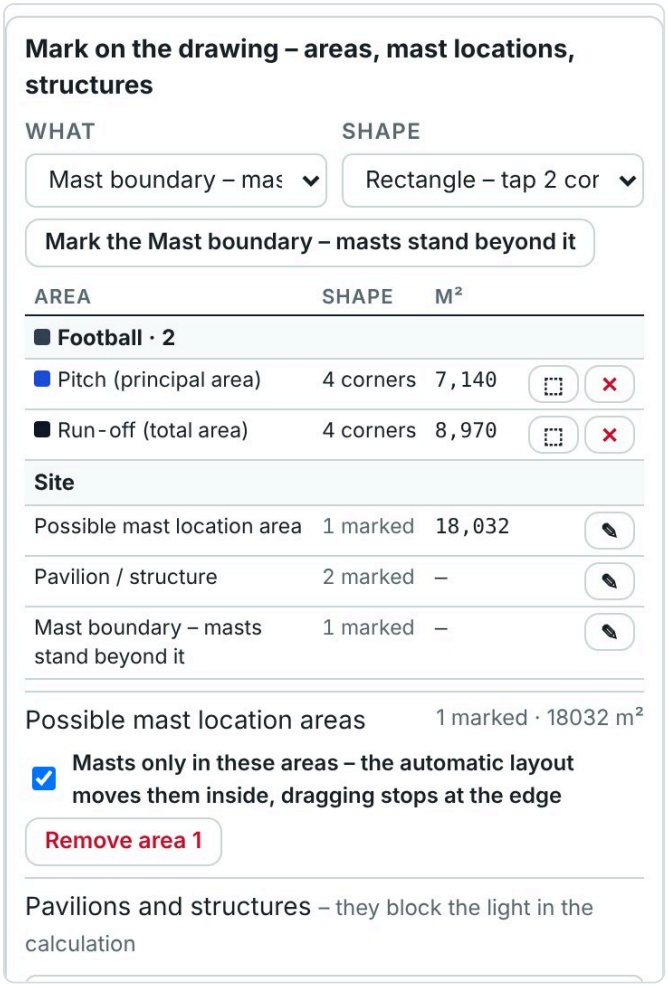
Marking the PDF drawing and designing



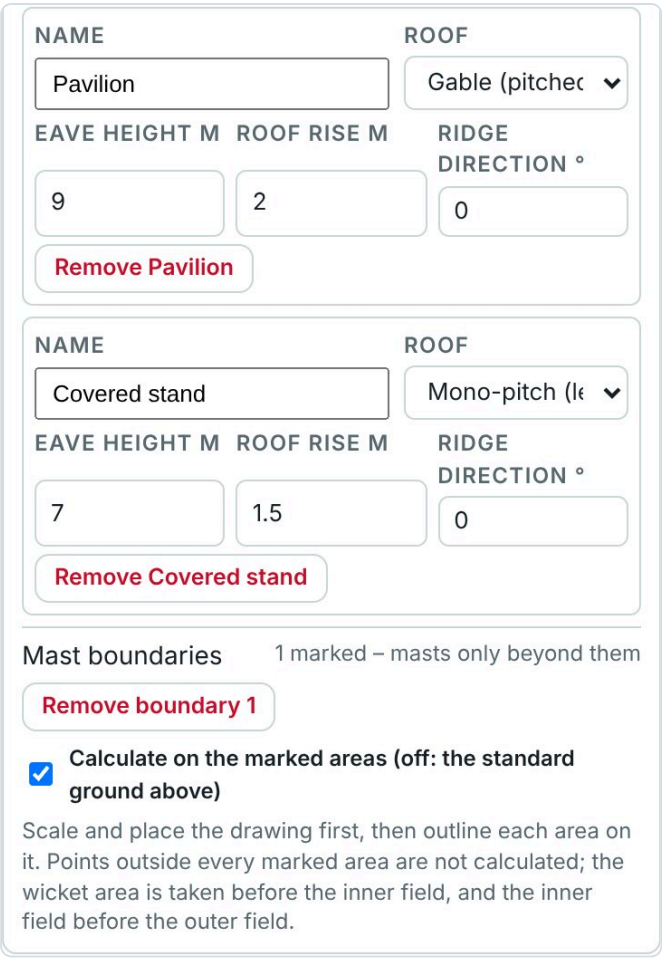
Mark on the drawing. Marking the pitch, rectangle shape.

4 · Mark the areas

1. **What:** Pitch (principal area). **Shape:** Rectangle – tap 2 corners. Press **Mark the Pitch** and tap two opposite corners on the drawing → 7,140 m².
2. The same for **Run-off (total area)** → 8,970 m².
3. **Possible mast location area**, outline: tap round the land the client allows for masts and press **Finish the outline** → 18,032 m². Tick **Masts only in these areas**.
4. **Pavilion / structure**, rectangle: the pavilion (gable roof, eaves 9 m, rise 2 m) and the covered stand (mono-pitch, 7 m, rise 1.5 m). They now block the light.
5. **Mast boundary** – rectangle round the playing area: masts stand beyond it.
6. Tick **Calculate on the marked areas**.

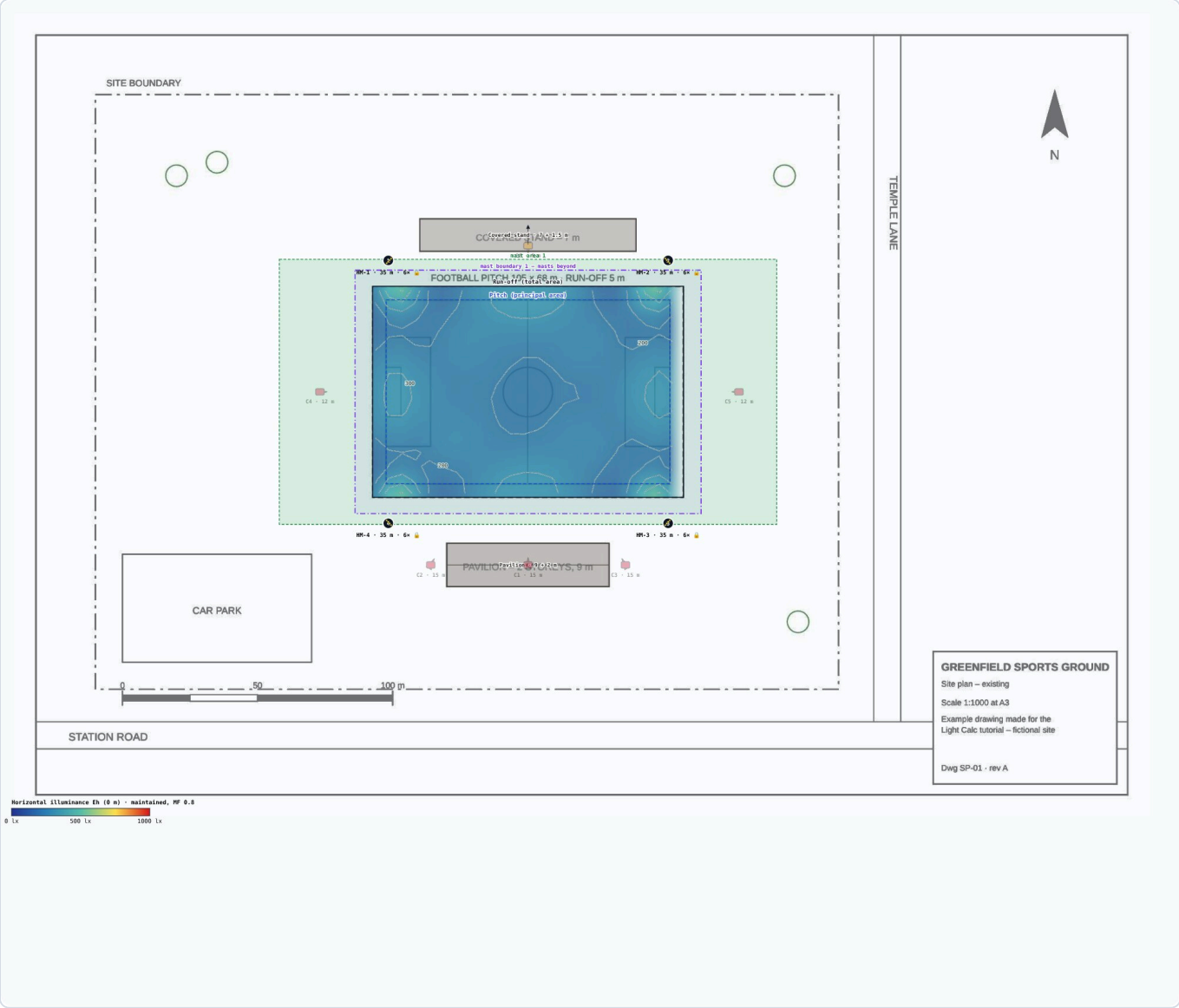


Everything marked. Shape and area for each; ⬚ edits the outline, ✕ removes it, ✎ marks another.



Structures. Name, roof type, eave height, roof rise, ridge direction.

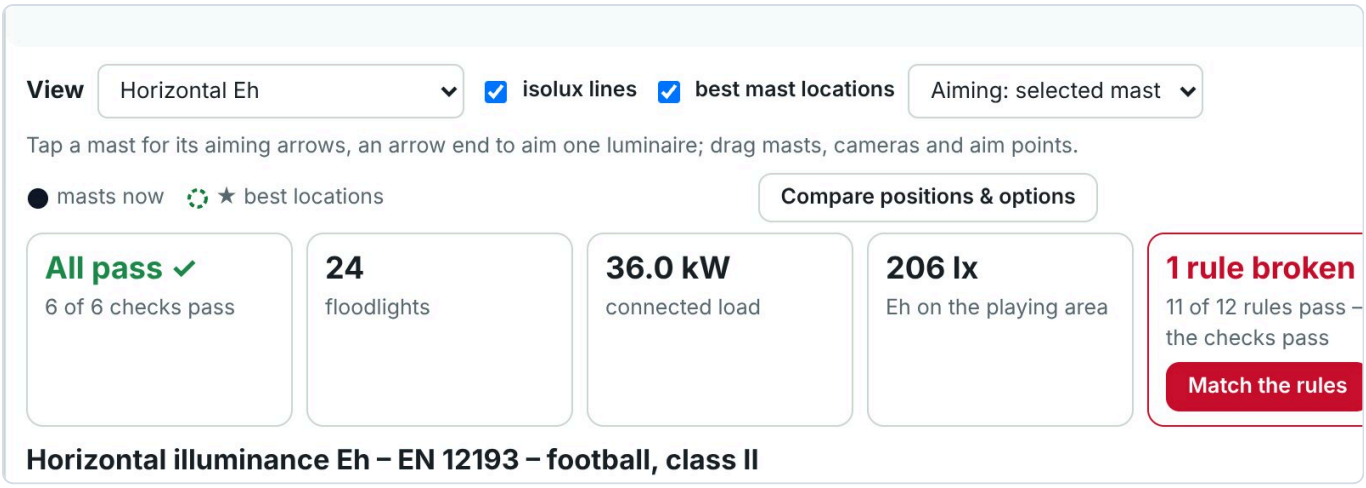
The result on the drawing



All marked. Pitch and run-off in blue, the mast location area in green, the pavilion and stand grey, calculated on the marked areas.

5 · Complete the design automatically

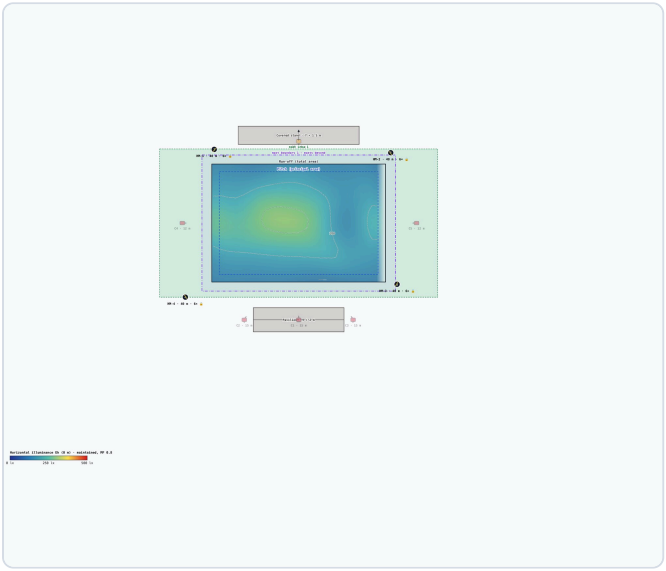
Mast locations **May change**, 4 masts. The placement keeps every mast inside the possible-location area and beyond the boundary.



Result. All 6 checks pass: 24 floodlights, 36.0 kW, 206 lx on the pitch, U2 0.63, run-off 145 lx, U2 0.70.

1 RULE BROKEN – AND WHY

The allowed land is not symmetrical about the pitch, so the masts cannot be either. **Match the rules** would move them out of the allowed area. Here the site wins: accept the warning, and the report states the reason.



Show the drawing unticked. The design stays on the marked areas and structures.

What the drawing adds

- Masts only where the client allows – no surprises at the site visit.
- Real shadows from the pavilion and the stand.
- The drawing goes into the report and the DXF export under the design.

CHAPTER 30

Worked case – an aerial picture or Google Maps screenshot

No drawing at all – only a satellite view. Here: *Riverside ground*, a cricket ground with a pavilion, taken as a 1600 × 1100 px screenshot with the map's 100 m scale bar.

TAKING THE SCREENSHOT

In Google Maps (satellite) or Google Earth, use the flat top-down view (no 3D tilt), north up, zoomed so the whole ground and its surroundings fill the window and the **scale bar** shows. Save it as PNG or JPG. Check the imagery terms of use for your project; the picture in this tutorial is drawn for the example.



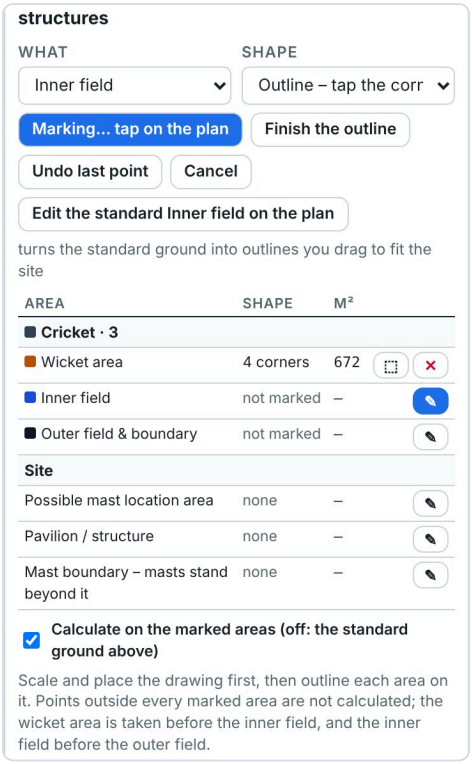
1 · Loaded. Sport Cricket first, then **Choose File**.



2 · Scale. The two taps on the scale bar (blue) and 100 m → 0.25 m per pixel.

3 · Move the picture so the centre of the pitch is on the plan's 0, 0 – rotate it first if the pitch is not along x. **No scale bar?** Use the pitch: 20.12 m between the stumps, or the boundary rope's radius if known.

4 · Mark with outlines



Outline in progress. Finish the outline, Undo last point, Cancel.

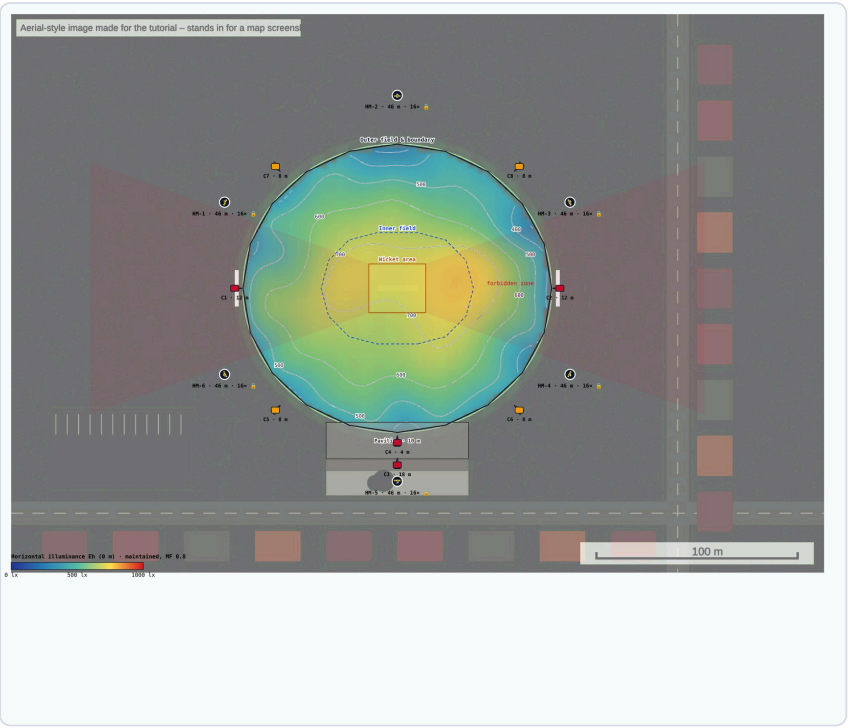
1. **Wicket area** – rectangle round the square → 672 m².
2. **Inner field** – outline: tap round the 30-yard circle; **Undo last point** for a mis-tap; **Finish the outline** → 14 corners, 3,361 m².
3. **Outer field & boundary** – outline along the rope → 20 corners, 16,671 m².
4. **Pavilion** – rectangle, flat roof, 10 m → 1,260 m²; it blocks the light.

The wicket area is taken before the inner field and the inner field before the outer field, so overlapping outlines are fine.



Tapping the inner field's corners (red points). The red wedges are the forbidden zones behind the wickets.

The aerial picture, marked and designed



Marked. Calculated on the three marked areas only.

Mark on the drawing – areas, mast locations, structures

WHAT: Pavilion / structure — SHAPE: Rectangle – tap 2 cor

Draw a pavilion / structure

AREA	SHAPE	M²
Cricket · 3		
Wicket area	4 corners	672
Inner field	14 corners	3,361
Outer field & boundary	20 corners	16,671
Site		
Possible mast location area	none	–
Pavilion / structure	1 marked	–
Mast boundary – masts stand beyond it	none	–

Pavilions and structures – they block the light in the calculation

NAME: Pavilion — ROOF: Flat roof

EAVE HEIGHT M: 10 — ROOF RISE M: 0 — RIDGE DIRECTION °: 0

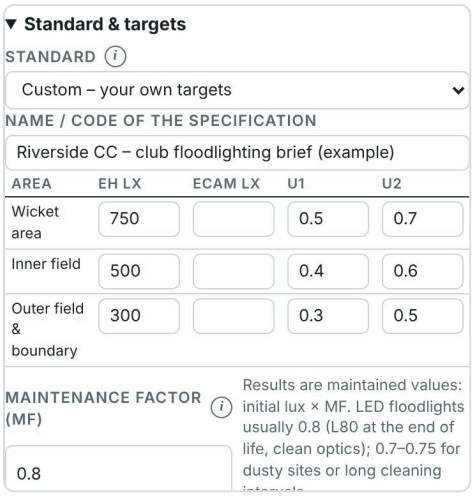
Remove Pavilion

☑ **Calculate on the marked areas (off: the standard ground above)**

Scale and place the drawing first, then outline each area on it. Points outside every marked area are not calculated; the wicket area is taken before the inner field, and the inner field before the outer field.

The list. Wicket 4 corners, inner 14, outer 20; one pavilion.

5 · The targets – a club brief on Custom



Custom – your own targets, named after the client's brief.

Riverside is a club ground with no television, so the international LiDAC targets (3000 / 2500 / 2000 lx with camera checks) would be wrong for it. Under **Standard & targets**, pick **Custom – your own targets**, name the specification and type its values. A blank box is not checked.

AREA	EH	U1	U2
Wicket area	750 lx	0.5	0.7
Inner field	500 lx	0.4	0.6
Outer field	300 lx	0.3	0.5

Example values for the tutorial – use the client's or the board's figures. Untick the cameras under **Cameras** when there is no broadcast.

TRACE ALONG THE ROPE

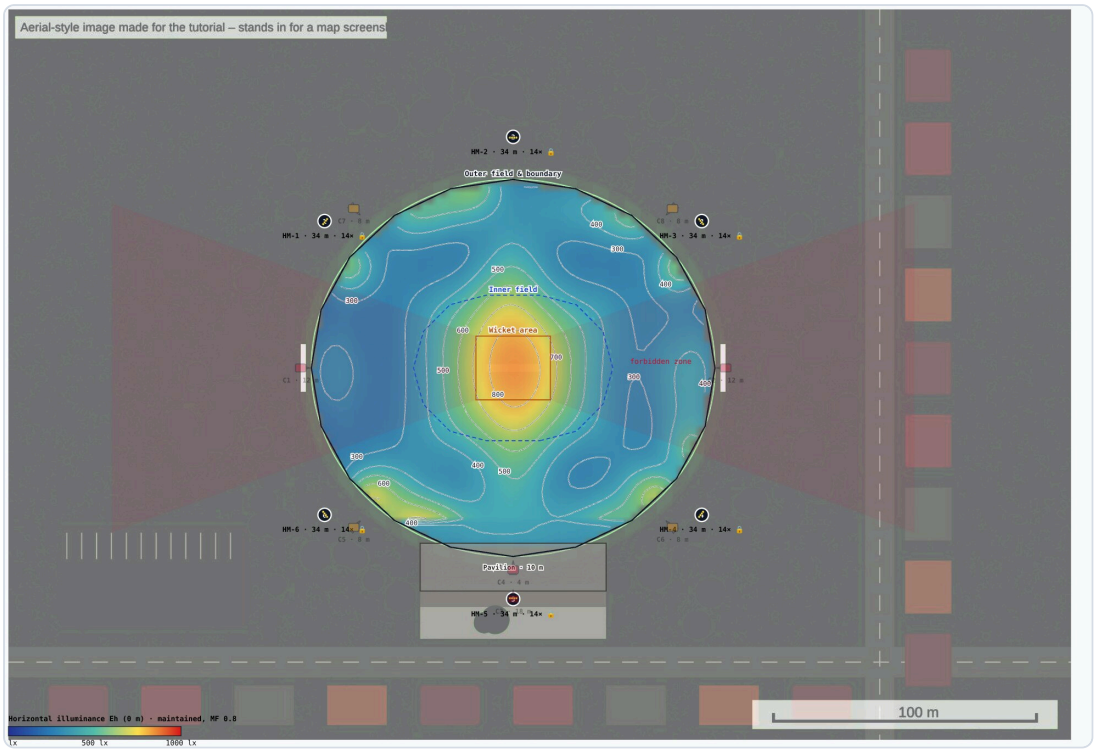
Draw the outer field's outline along the boundary rope, not over the pavilion. Points beyond the outline are not calculated; from v156 the ground inside a closed building is left out as well.

THE SAME GROUND AT INTERNATIONAL TV LEVEL

With LiDAC international targets the same six masts needed about 684 floodlights (about 1 MW) and still missed the camera uniformity – the tool says to use 8 masts. The class decides the size of the job: check it first.

The cricket ground designed

6 · Complete the design automatically – mast locations **May change**, 6 masts (the LiDAC minimum). 107 seconds later:



The aerial picture designed. Six 34 m masts 15 m beyond the boundary, end masts 39.5° from the pitch axis and out of the forbidden zones behind the wickets; the pavilion casts its shadow.

All pass ✓
9 of 9 checks pass

84
floodlights

126.0 kW
connected load

808 lx
Eh on the wicket

Rules ✓
11 of 11 rules pass

Beam angles
84 × ~25°
Suggest beam angles

Horizontal illuminance Eh – Riverside CC – club floodlighting brief (example)

AREA	EH AVERAGE	MIN / MAX	U1 MIN/MAX	U2 MIN/AVE	TARGET EH · U1 · U2
Wicket area	✓ 808 lx	708 / 868	✓ 0.82	✓ 0.88	750 lx · 0.5 · 0.7
Inner field	✓ 583 lx	356 / 817	✓ 0.44	✓ 0.61	500 lx · 0.4 · 0.6
Outer field & boundary	✓ 374 lx	235 / 705	✓ 0.33	✓ 0.63	300 lx · 0.3 · 0.5

All 9 checks and all 11 rules pass – 84 floodlights, 126 kW; wicket 808 lx (U2 0.88), inner field 583 lx (0.61), outer field 374 lx (0.63).

✓ 6 masts (minimum 6 – fewer give a weak edge and harsh shadows)
✓ None in the forbidden zone behind the wickets
✓ All at least 5 m beyond the boundary
✓ All masts high enough for aiming within 70°
✓ All masts at least 41 m apart
✓ No mast on a structure
✓ Layout symmetrical for play in both directions
✓ All aiming within 70° of vertical
✓ All aim points on the calculation areas (no beam aimed at the spectators)
✓ Glare GR Batsman, end A: 30 (CIE: < 40)

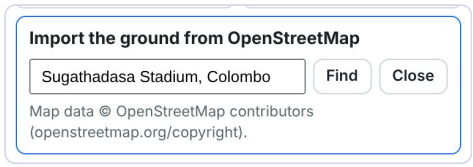
The cricket rules, including no mast on a structure and the batsmen's glare at each end (GR 30 and 33, below 40).

CHAPTER 31

Importing the ground from OpenStreetMap

When the ground is mapped on OpenStreetMap, the outlines come in already drawn and to scale – no tapping.

1. Under **Ground, grid & drawing** press **Import the ground from OpenStreetMap**.
2. Type the ground's name and town, or *latitude, longitude* (from Google Maps: right-click → the numbers), and press **Find**.
3. If several places match, pick one from the list.
4. Choose the **sport** (fixed once imported).
5. A table lists every pitch, track and stadium outline within 600 m – what it is on the map and its size. Under **Use as**, say what each one is: playing area / pitch, run-off, ground outline, or leave it out.
6. Tick **Put the map under the plan** for the map tiles as the drawing.
7. Press **Import these outlines**.



The OpenStreetMap pane, a ground typed in.

What the import does

- The plan is turned so the pitch (cricket) or playing area runs along x; the map turns with it.
- **Cricket:** the wicket area and the 30-yard circle are set round the cricket pitch. **Other sports:** an unmarked run-off becomes the standard band round the pitch.
- **Multi-sport:** every outline becomes a sport with its standard and standard run-off, or a measurement area.
- Every area can be moved or redrawn afterwards, as in chapter 32.
- Later, under *Neighbours & obtrusive light*, **Find buildings (OpenStreetMap)** places a window on the nearest façade of each building round the ground (chapter 17).

WHEN IT DOES NOT WORK

No pitches, tracks or stadiums are mapped within 600 m – the ground is not on OpenStreetMap yet: use a drawing or an aerial picture (chapters 29–30). *Servers busy or blocked* – four outline servers are tried; if none answers on your network, ask IT to allow **nominatim.openstreetmap.org**, **overpass-api.de** and **tile.openstreetmap.org**. Map data © OpenStreetMap contributors.

ABOUT THIS CHAPTER

The network these screens were taken on blocks OpenStreetMap, so no imported result is shown here. The same ground marked by hand is the worked case in chapter 30.

CHAPTER 32

Marking areas – every option

WHAT	EFFECT
Pitch / court / infield / wicket area (principal area)	Replaces the standard principal area – calculated and checked against its targets.
Run-off / track / inner & outer field (total area)	Replaces the standard total area.
Possible mast location area	Land where masts may stand. With **Masts only in these areas** ticked, the automatic layout moves masts inside and dragging stops at the edge. Several can be marked.
Pavilion / structure	A building that blocks the light: name, roof (flat, gable, hipped, mono-pitch…), eave height, roof rise, ridge direction. The ground inside it is not calculated, so an area outline may run over it (open canopies excepted).
Mast boundary	Masts must stand beyond it (outside it).

SHAPE	HOW
Outline – tap the corners	Tap each corner; **Undo last point**; **Finish the outline** (at least 3 points).
Rectangle – tap 2 corners	Two opposite corners, square to the plan.
Circle – centre, then edge	For a round ground, a shot circle or a roundabout of land.

Buttons in the list

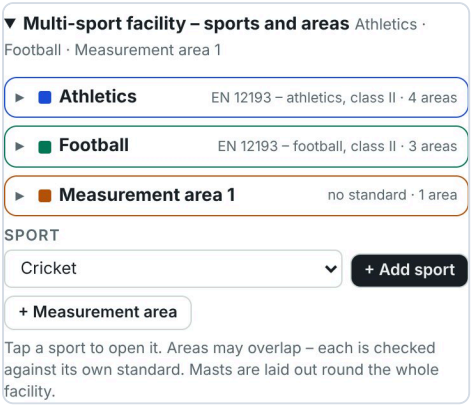
- ✎ mark this area (or one more of a site item) on the drawing.
- ⬚ edit the outline on the plan: drag its corners.
- ✕ remove the marked outline – the standard area comes back.
- **Edit the standard … on the plan** turns the standard ground into outlines you can drag to fit the site, without a drawing.
- **Calculate on the marked areas** – off: the standard ground above is used. Points outside every marked area are not calculated.

TIP

No drawing at all? **Edit the standard pitch on the plan** and drag its corners to the surveyed sizes – the quickest way to fit an odd-shaped ground.

CHAPTER 33

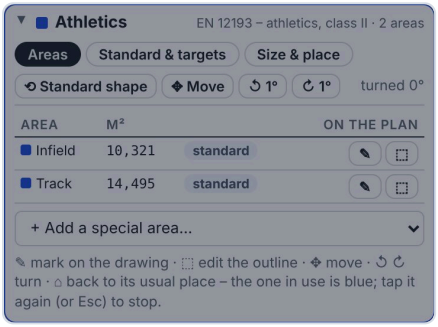
The multi-sport panel



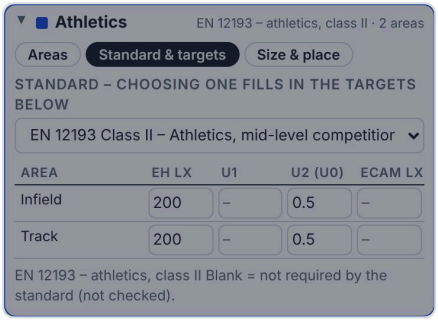
Multi-sport facility – sports and areas.

Choose **Multi-sport facility** as the sport. Pick a sport under **Sport** and press **+ Add sport** – it is placed with its standard shape and standard. **+ Measurement area** adds a free area with your own targets (a car park, a warm-up area). Areas may overlap – each is checked against its own standard; masts are laid out round the whole facility.

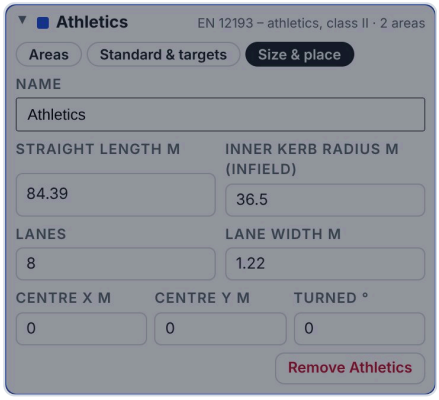
Tap a sport to open its three tabs.



Areas. ⟲ Standard shape, ✥ Move, ↺ 1° / ↻ 1° turn, ✎ mark on the drawing, ⬚ edit the outline, **+ Add a special area...**



Standard & targets. The standard fills the targets – Eh, U1, U2, Ecam – for each area. Blank = not required.



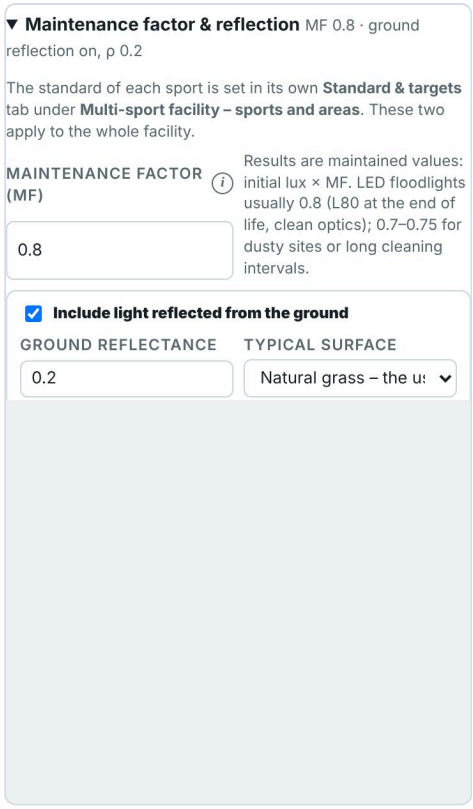
Size & place. Name, the sport's sizes, centre x / y, turned °; **Remove** the sport.

Special areas

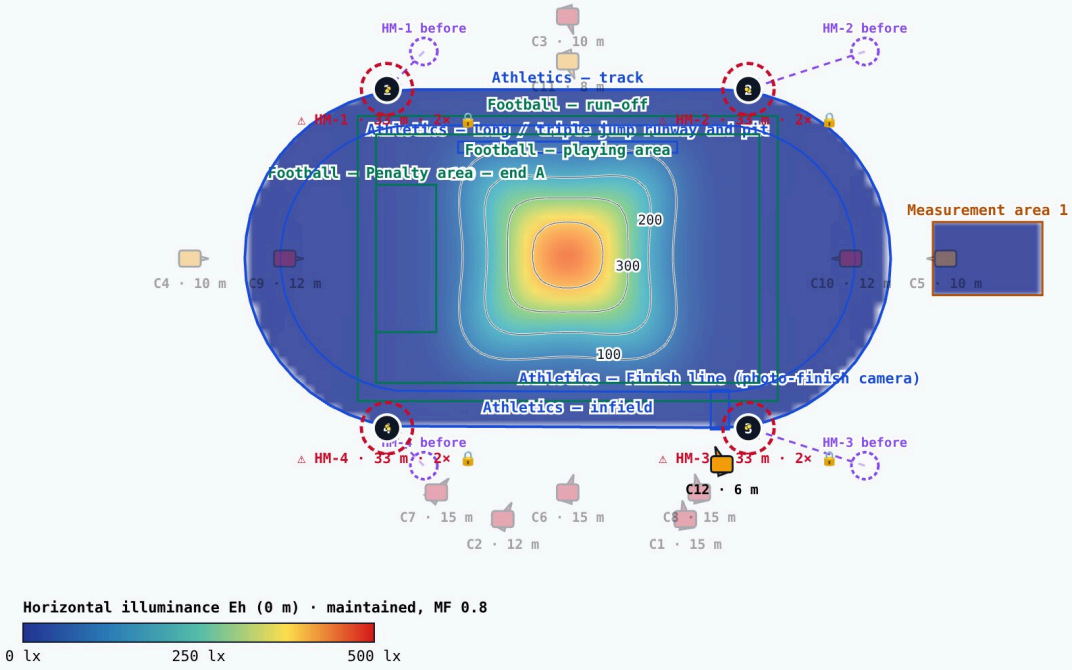
Under **Areas**, **+ Add a special area…** lists the sport's own: the athletics finish line, sprint straight, jumps and throws; football penalty areas, goal areas and centre circle; rugby in-goals and 22 m areas; the cricket strip. Each is placed at its standard position with its own target and checked separately – for example the finish line for the photo-finish camera.

Maintenance factor & reflection

For a multi-sport facility, MF and the ground reflectance apply to the whole site, in their own pane. Each sport's standard stays in its own tab.



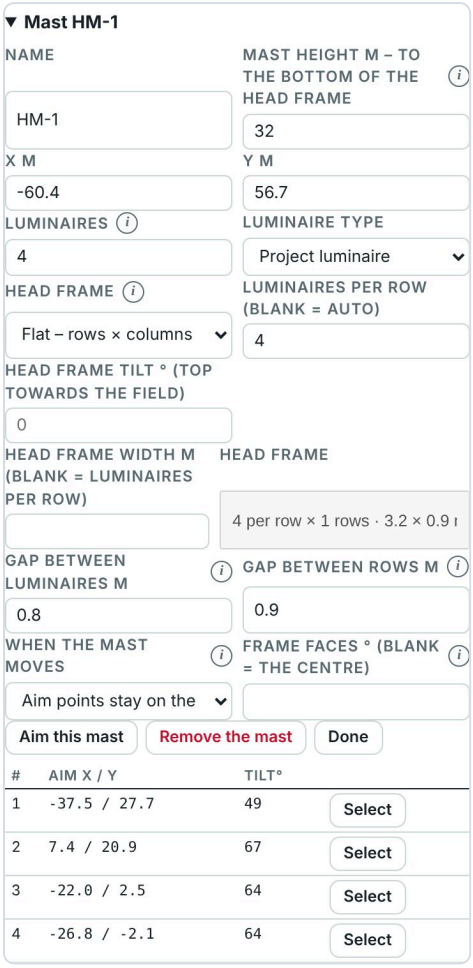
MF & reflection for the whole facility.



On the plan. Athletics track and infield, football pitch and run-off, the finish line and a measurement area, each labelled. “before” circles show where the masts stood before the last run.

CHAPTER 34

The mast editor



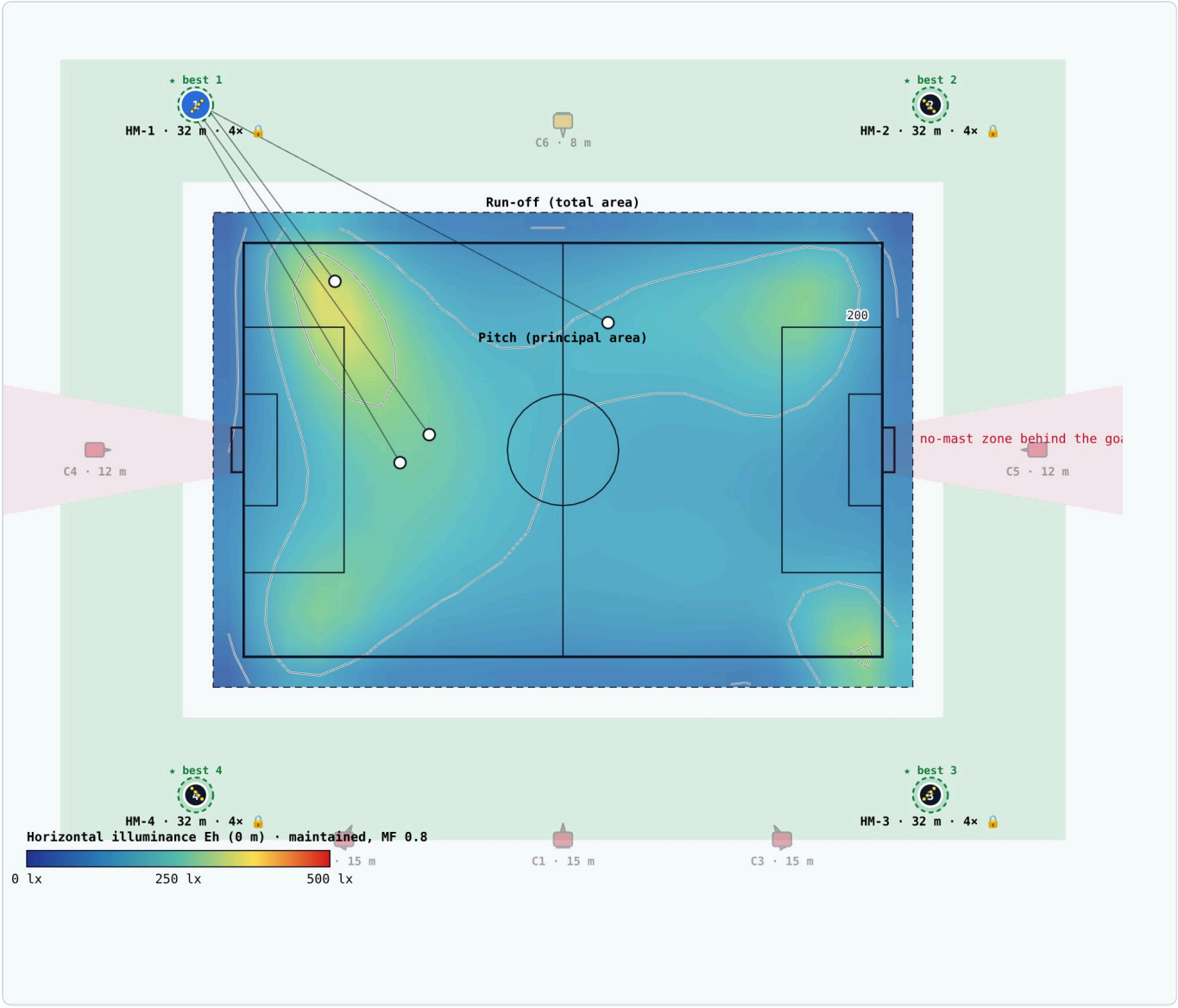
Mast HM-1. Tap a mast on the plan to open it.

FIELD	MEANING
Name	HM-1… – used in the report, aiming PDF and cable plan.
Mast height	To the bottom of the head frame.
x / y m	Position from the centre. Or drag the mast on the plan.
Luminaires	Number on this mast; new ones copy the existing aim points.
Luminaire type	The project luminaire or any loaded IES / LDT – mix types per mast.
Head frame	Flat rows × columns, stepped, curved, V, trapezoid or ring.
Luminaires per row	Blank = automatic.
Head frame tilt °	Top towards the field.
Head frame width	Blank = from luminaires per row; the frame size shows beside it.
Gap between luminaires / rows	Centre to centre – at least the luminaire size plus room to aim.
When the mast moves	Aim points stay on the field (locked) · aiming moves with the mast · re-aim automatically.
Frame faces °	0 = along +x. Blank = towards the centre.

Aim this mast aims only this mast's floodlights. **Remove the mast**. **Done** closes it.

The aim table

Each floodlight's aim point x / y and tilt. **Select** highlights its aiming line on the plan; drag its end to re-aim.



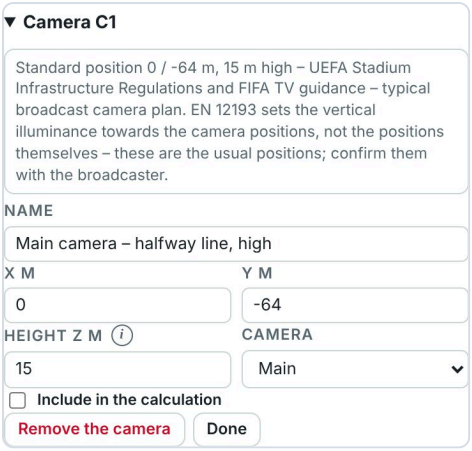
The selected mast with its aiming lines, the ★ best locations and the no-mast zones behind the goals.

SINGLE FLOODLIGHTS

A floodlight has no separate pane: select it in the aim table and drag its aim point on the plan.

CHAPTER 35

The camera editor



Camera C1. Tap a camera on the plan.

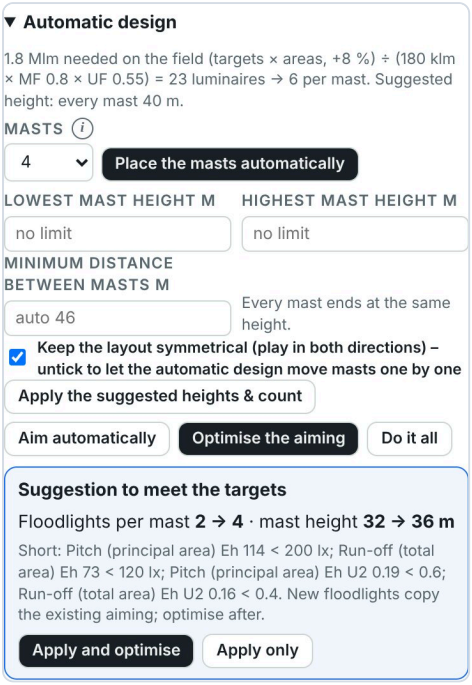
The grey box gives the standard position for this camera and its source – here *0 / −64 m, 15 m high, UEFA Stadium Infrastructure Regulations and FIFA TV guidance*. EN 12193 sets the vertical illuminance towards the cameras, not where they stand, so confirm the positions with the broadcaster.

FIELD	MEANING
Name	For example “Main camera – halfway line, high”.
x / y m	Position; or drag the camera on the plan.
Height z m	Lens above the field.
Camera	Main or secondary – the targets differ.
Include in the calculation	Ticked: Ecam and uniformity towards this camera are calculated and checked.

Remove the camera; **Done**. Under **Cameras**: **+ Camera** adds one; **Standard cameras** places the usual set for the sport.

CHAPTER 36

Design panes: suggestion, rules, groups and electrical



Suggestion to meet the targets.

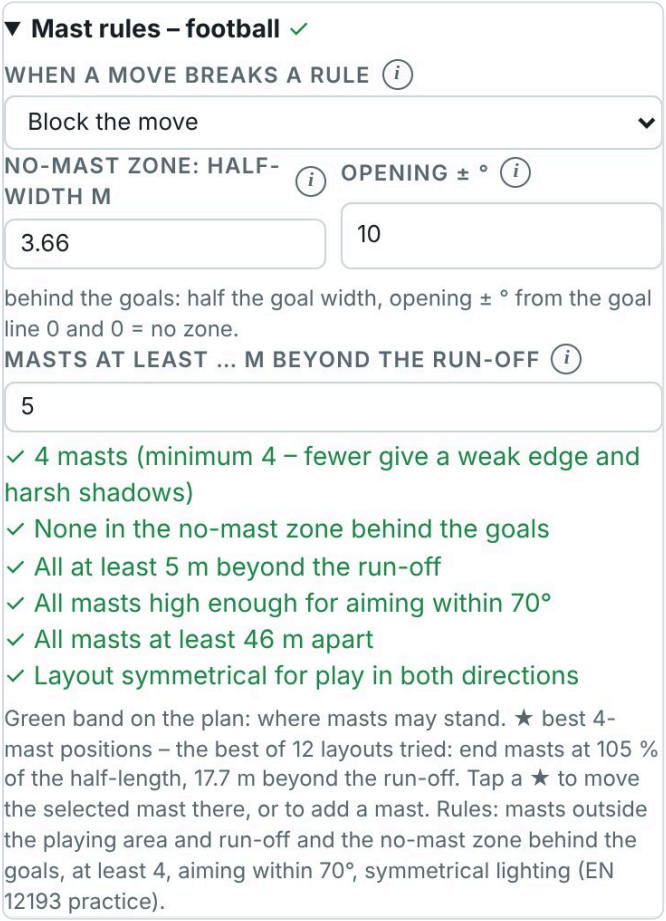
The suggestion box

When the design falls short, the **Automatic design** pane shows what to change, with the reason: here floodlights per mast 2 → 4 and mast height 32 → 36 m, because the pitch has 114 lx against 200 lx and U2 0.19 against 0.6. **Apply and optimise** applies it and re-aims; **Apply only** applies it without aiming.

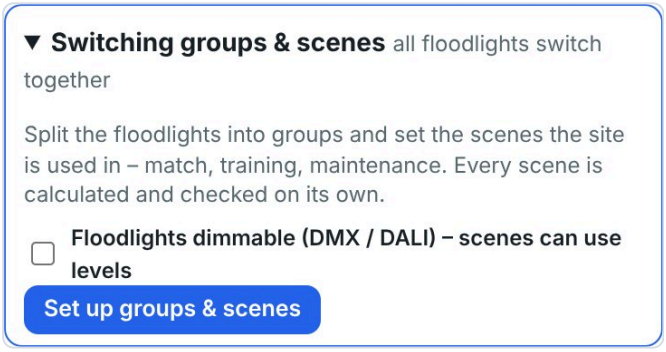
The line on top is the lumen method: lumens needed on the field ÷ (luminaire lumens × MF × UF) = floodlights, and the suggested height.

Step by step instead of "Do it all"

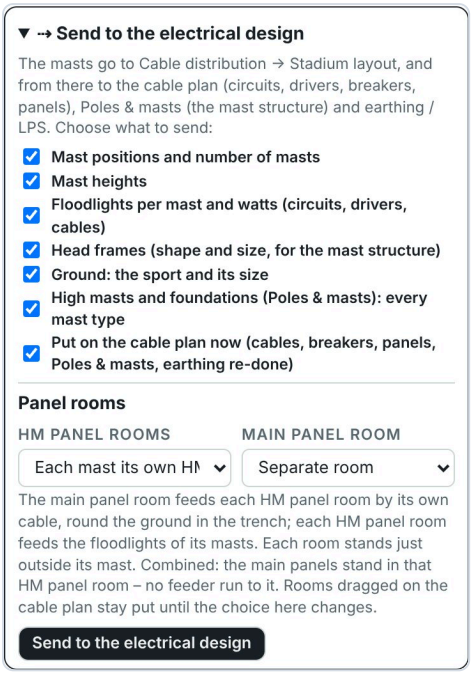
Place the masts automatically · **Apply the suggested heights & count** · **Aim automatically** · **Optimise the aiming** · **Do it all**. Lowest / highest mast height and minimum distance between masts limit the search; **Keep the layout symmetrical** can be unticked to let the design move masts one by one.



Mast rules. When a move breaks a rule (block or allow), the no-mast zone, masts beyond the run-off; every rule listed ✓ or ⚠.



Switching groups & scenes before set-up: tick **Floodlights dimmable** for levels, then **Set up groups & scenes** (chapter 12).



Send to the electrical design.

Send to the electrical design

The masts go to **Cable distribution → Stadium layout**, from there to the cable plan, Poles & masts and earthing. Tick what to send: mast positions and number, mast heights, floodlights per mast and watts, head frames, the ground, high masts and foundations, and **Put on the cable plan now**.

HM panel rooms: each mast its own, or 2 (or 4) shared. **Main panel room:** a separate room, or with one of the HM rooms. Then press **Send to the electrical design**. The choice is kept until changed here.

APPENDIX A

Limits and targets used

Obtrusive light – EN 12464-2 table 2 / CIE 150

ZONE	EV ON WINDOWS (BEFORE / AFTER CURFEW)	I PER FLOODLIGHT	ULR	TI (LAV)
E1 natural	2 / 0 lx	2,500 / 0 cd	0 %	15 % (0.1 cd/m²)
E2 rural	5 / 1 lx	7,500 / 500 cd	5 %	15 % (1)
E3 suburban	10 / 2 lx	10,000 / 1,000 cd	15 %	15 % (2)
E4 city centre	25 / 5 lx	25,000 / 2,500 cd	25 %	20 % (5)

The planning authority may set its own figures – type them under *Limits from the planning authority*; the report then says so.

Guidance values (editable, not standard requirements)

- Crowd shots: faces to the main camera 25 % of the pitch's camera illuminance.
- Colour and flicker: as in chapter 19.
- Run-off: 60 % of the playing-area value as a design allowance where the standard sets none.

Mast rules – football (defaults)

At least 4 masts; none behind the goals within the goal width ± the opening; at least 5 m beyond the run-off; aiming within 70° of vertical; masts at least the minimum distance apart; layout symmetrical for play in both directions.

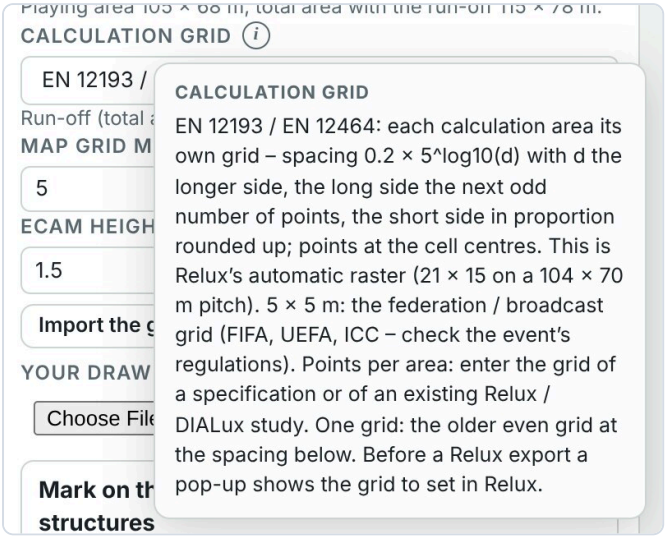
Appendix B – Glossary

TERM	MEANING
Eh	Horizontal illuminance on the playing surface (lx)
Ecam, Ev	Illuminance on a plane facing a camera / a window (lx)
U1, U2	Uniformity: Emin / Emax and Emin / Eave
MF	Maintenance factor – maintained = initial × MF
GR	Glare rating, CIE 112 (lower is better; ≤ 50 for players in EN 12193)
ULR	Upward light ratio – share of the floodlights' flux above the horizontal
TI	Threshold increment – glare for drivers
CCT, Ra, R9, TLCI	Colour temperature; colour rendering index and its saturated-red sample; television lighting consistency index
Scene	A set of group levels checked against a class – match, training, televised, security
Run	A length of roof catwalk with its floodlights, designed like a mast
Must / should / report	A scene must pass to finish the design / is aimed for / is only reported
.rdf	Relux project file – exported and imported

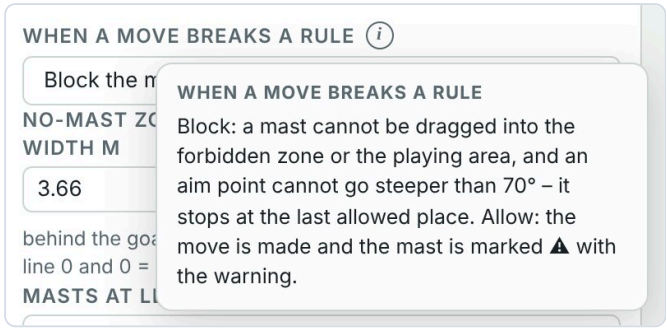
APPENDIX C

Every “i” help text

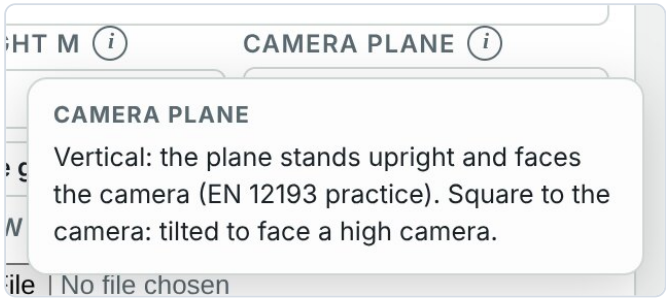
Tap the **i** beside a field for what to enter. Tap it again, or anywhere else, to close.



The help for **Calculation grid**.



The help for **When a move breaks a rule**.



The help for **Camera plane**.

FIELD · PANE	HELP TEXT
Standard Standard & targets	The specification the design is checked against. Each one fills the targets for the three areas; change any value for the project, or pick Custom and name the client's specification.
Boundary half-length Ground (cricket)	Cricket boundary, measured from the centre of the pitch along the pitch (59–82 m for international grounds).
Boundary half-width Ground (cricket)	Boundary measured across the pitch. Equal to the half-length for a round ground.
Wicket area half-length Ground (cricket)	Half-length of the wicket area (the ellipse round the pitch) – the zone that needs the highest and most even light.
Wicket area half-width Ground (cricket)	Half-width of the wicket area.
30-yard circle radius Ground (cricket)	The 30-yard (27.43 m) circle: two half-circles round the middle stumps joined by straight lines – the inner field.
Forbidden zone ± ° Mast rules (cricket)	No mast may stand in line with the pitch behind the wickets – the batsman and bowler would look into the lights. The zone opens at ± this angle from the edges of the wicket area, so it covers any pitch laid in it.
Masts at least … m beyond the boundary Mast rules	Masts must stand outside the playing area – this much beyond the boundary rope, for the boundary run-off, advertising boards and the sight screens.
When a move breaks a rule Mast rules	Block: a mast cannot be dragged into the forbidden zone or the playing area, and an aim point cannot go steeper than 70° – it stops at the last allowed place. Allow: the move is made and the mast is marked ⚠ with the warning.
Masts from the stadium layout Masts	Ticked: the masts here are the stadium layout's from Cable distribution – positions, heights and floodlights come across whenever the layout changes there. Untick to move masts freely here.
Wicket area shape Ground (cricket)	The wicket area round the pitch, where the highest and most even light is needed. A rectangle (or a square with equal sizes) is the usual calculation area; the ellipse follows the LiDAC sketch.
Maintenance factor (MF) Standard & targets / MF pane	Maintenance factor: the share of the new light left at the end of the maintenance period (LED ageing, dirt on the glass). Every result is maintained = initial × MF. LED floodlights: 0.8 is usual; 0.7–0.75 on dusty sites.
Masts (number) Automatic design	Number of masts for the automatic placement – at least 6 (LiDAC). The search tries the end masts at several angles from the pitch axis and several distances beyond the boundary, aims each layout and keeps the one closest to the targets.
Grid spacing Ground, grid & drawing	Per area (default): each area is calculated on its own grid like Relux – points at the cell centres, spacing 0.2 × 5^log d (EN 12464), so the figures match a Relux calculation; this spacing is then only for the colour map. One grid: every figure from one even grid at this spacing. The wicket area always uses a finer grid (≤ 1.5 m).
Calculation grid Ground, grid & drawing	EN 12193 / EN 12464: each calculation area its own grid – spacing 0.2 × 5^log10(d) with d the longer side, the long side the next odd number of points, the short side in proportion rounded up; points at the cell centres. This is Relux's automatic raster (21 × 15 on a 104 × 70 m pitch). 5 × 5 m: the federation / broadcast grid (FIFA, UEFA, ICC – check the event's regulations). Points per area: enter the grid of a specification or of an existing Relux / DIALux study. One grid: the older even grid at the spacing below. Before a Relux export a pop-up shows the grid to set in Relux.
Ecam height Ground, grid & drawing	Height of the vertical calculation points for the cameras – 1.5 m in the specification (the height of the players and the ball).
Camera plane Ground, grid & drawing	Vertical: the plane stands upright and faces the camera (EN 12193 practice). Square to the camera: tilted to face a high camera.
Mast height Mast editor	Height of the luminaires (the head frame) above the field. The suggestion meets 25° from the centre of the pitch and keeps aiming within 70° of vertical.

When the mast moves Mast editor	Locked: the aim points stay where they are on the field when you drag the mast. Moves with the mast: the aiming travels with it. Re-aim: aimed again automatically after the move.
Head frame Mast editor	How the luminaires sit on the head frame: rows × columns on a flat frame, rows stepped back, or round a ring.
Gap between luminaires Mast editor	Centre-to-centre gap between luminaires in a row – at least the luminaire width plus room for aiming.
Gap between rows Mast editor	Vertical gap between the rows of luminaires.
Luminaires Mast editor	Number of floodlights on this mast. More keep the aiming: new ones copy the existing aim points.
Height of all masts Heights & floodlights	Sets every mast to this height. Blank when the masts differ. The aim points stay where they are.
Floodlights on every mast Heights & floodlights	Sets the number of floodlights on every mast. The aiming is kept – new floodlights copy the existing aim points; optimise after.
Frame faces ° Mast editor	Direction the head frame faces, in degrees (0 = along +x). Blank: it faces the centre of the pitch.
Height z Camera editor	Height of the camera lens above the field.

APPENDIX D

Control index

Where each control is explained.

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